

Liquid Markets: An Empirical Analysis of a Water Exchange

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Abstract

This paper studies how the trading mechanism affects water market outcomes, by structurally analyzing exchange data from Australia, the world leader in water trade. In this novel application of multi-unit exchange, which resembles a limit order market, I combine data on buy and sell orders and trades with models of participants' best-response actions to identify and estimate their underlying willingness to pay. I also extend my baseline model for robustness, endogenizing order quantity in addition to price and incorporating behavioral flexibility. Counterfactual simulations with the estimated models reveal there is ample room to improve water exchange through design, design matters in an absolute and relative sense, and the impact is especially pronounced during drought, when improvements are most critical. I decompose how different exchange characteristics—including markups, continuous trading versus batching, and (in)divisibility of parcels—contribute to the performance gap and propose a design reform eyeing both effectiveness and practicality.

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1 Introduction

According to UN-Water, which coordinates the United Nations’ work on water and sanitation, water is the primary medium through which we will feel the effects of climate change. As a result, there is growing interest in water reform, particularly in the potential of water markets to help overcome these problems.¹ This paper investigates one piece of the broader puzzle: how the design of the trading mechanism, or market microstructure, impacts water market outcomes such as the gains from trade and how these impacts interact with drought. I approach this question through a structural analysis of water exchange data.

I draw on exchange data from Australia, the driest inhabited continent and world leader in market-based water management. Australia’s advanced markets feature a zone-based water trading system that commoditizes water within each zone and online exchanges that obviate costly search by collecting posted price-quantity offers from individual traders. The study of exchange in Australian water is distinctly valuable because other countries look to Australian water markets as a model example; water markets elsewhere are largely in pre-nascent stages. The Wall Street Journal reports that officials from China, India, Europe, Japan, and Egypt visited Australia to examine the market’s operation.² Exchange features present in Australia are likely to influence future exchange elsewhere, and lessons drawn regarding their merits and flaws have significance beyond the case study at hand. Beyond water, the lessons are relevant for other nascent exchanges that could emerge in similar fashion as more environmental problems seek market solutions.

In this novel application of multi-unit exchange, I find that there is a lot of room to improve the market through exchange design, as measured by the gap in trade surplus between the status quo and a performance benchmark. This assessment is robust to different models of trader behavior and different geographic areas. I find that the design of how trades are arranged has an important impact, not only in an absolute sense but also relative to a major liquidity expansion event, and the impact is especially pronounced during drought, when improvements are most critical. Therefore, I search for and propose a design reform with an eye to both effectiveness and real-world practicality. To this end, I decompose how different exchange characteristics—including

¹National Water Commission (2011), Australia.

²See Taylor (2018).

markups in posted prices, continuous trading versus batching, and (in)divisibility of water parcels—contribute to the performance gap. A parcel refers to the water supplied or demanded by a trader in a single order, comprising multiple units (megaliters) each. The existing transaction fee structure incentivizes parcel-level, as opposed to unit-level, allocation, contributing to a type of indivisibility problem. I find that this interferes substantially with the efficient allocation of water units and presents low-hanging fruit for increasing the gains from trade.

The spot market for water in Australia, particularly in the southern part of its Murray-Darling river basin (southern MDB), plays a significant role in agriculture. In 2018-2019, the average dairy, horticulture, and rice farm relied on “allocation trades,” which refer to spot trades of water, for 41%, 39%, and 23% of their water use, respectively.³ One of the main media for allocation trades in Australia are online exchanges, which resemble but also differ from financial limit order markets. To trade, a buyer or seller (trader) places either a limit order or a market order. Taking the buyer’s perspective for expositional clarity, placing a limit order means the buyer lists the price-per-unit and quantity at which they want to buy, in the uncertain hope that a willing seller arrives to fulfill those terms. Placing a market order means the buyer selects an existing seller’s limit order to fulfill, for guaranteed trade. There is no market maker; each distinct trade can happen at a different price from other trades occurring at the same time. Thus, the exchange features continuous trade at discriminatory prices. Due to imperfect divisibility of parcels, I observe that buyers effectively match with sellers’ parcels rather than buy the lowest-priced units from the aggregate supply curve. I also observe that limit orders with not very competitive prices still have a decent chance of becoming a trade, incentivizing traders to mark up or mark down their limit order prices from their true valuations. I refer to this as price shading.

To assess counterfactual exchange design as described above, I need to recover buyers’ latent willingness to pay (“values”) and sellers’ latent willingness to accept (“costs”) from observed data. Meanwhile, general limit order markets are complex dynamic games of which the solution is analytically intractable, and much is still unknown despite a long history of research in the finance literature.⁴ This paper does

³Australian Competition and Consumer Commission (2021). The remainder of farms’ water use would be supplied by the water licenses (“entitlements”) they own. See Section 2.1 about entitlements. Only a small proportion of irrigators across the Basin use more sophisticated derivative products such as leases or forward contracts.

⁴Per Parlour and Seppi (2008), “there is still much we do not know about limit order markets.

not provide a full equilibrium theory of limit order markets but focuses on deriving key identifying equations that exploit the data. The equations impose the minimum structure needed to identify only the model primitives necessary to simulate a set of counterfactual exchange designs. In this vein, I model a trader’s *best-response* action on the exchange given her value for water.

In the baseline, I model the trader’s choice between placing a limit order and a market order and, conditional on placing a limit order, the choice of a limit order price. Taking the buyer’s perspective, buy limit orders with higher prices are more likely to be executed because sellers prefer higher prices. Thus, a buyer’s optimal limit order price balances the marginal benefit of increased execution probability against the marginal cost of paying a higher price conditional on execution. The buyers’ optimization in light of this tradeoff generally leads them to shade their prices and leads buyers with higher values to list higher limit order prices. When it comes to the choice between market and limit orders, traders follow a threshold strategy: buyers place a market order if their value exceeds an indifference threshold and place a limit order otherwise, because the guarantee of trade offered by market orders is more valuable for higher-value buyers. I then extend the baseline model in new ways to allow endogenous choice of order quantities, risk aversion, and market norms or default choices. These accommodate considerable flexibility in order placement behavior.

The traders’ first-order necessary condition for limit order pricing, described above, is then the key to identifying buyers’ latent values and sellers’ latent costs from order data. Hollifield, Miller, and Sandås (2004) and Hollifield, Miller, Sandås, and Slive (2006)—to my knowledge the first and only existing structural analyses of limit order markets—exploit analogous tradeoffs to identify traders’ values on a stock exchange, though model details differ due to differences between stock markets and water markets. The first-order condition takes as an input the empirical probability of limit order execution as a function of limit order price, in the spirit of Guerre, Perrigne, and Vuong (2000). There are additional complexities to identifying the full model, such as those arising from traders’ selection into market versus limit orders; I address these to establish formal identification. In my extended model with endogenous order

[...] Only very stylized environments have been studied thus far. Joint decisions about order aggressiveness and quantity have not been fully modelled [...].” Roşu (2009) explains that “one of the reasons for the scarcity of models of order-driven markets is the sheer complexity of the problem.”

quantities, I show that the added information from observing each optimized order quantity further identifies the local demand curve of each buyer and the local supply curve of each seller, i.e., what their valuations would have been at counterfactual order quantities. I then implement an estimation procedure that closely aligns with these identification arguments. The procedure accounts for heterogeneity in the market environment by conditioning on a vector of descriptive covariates, such as rainfall and water stored in major lakes.

Aiming for flexibility, nuance, and robustness, I assess exchange performance using multiple models, time periods, and locations. Specifically, I present results under different utility function specifications (higher versus lower risk aversion) and for exogenous and endogenous order quantity models, subdivide the data sample into shorter time periods and report period-specific outcomes, and replicate the analysis for two different geographic zones to learn what lessons are externally valid.

The key data that enable this analysis are order data from a major exchange. Order data are critical to evaluating the workings of an exchange because the exchange takes orders as inputs to then yield trades as outputs. Not all orders become trades, so orders reveal information about supply and demand—and the exchange’s ability to arrange that supply and demand into trades—that cannot be recovered from realized trades alone. That said, I caveat that important questions concerning water markets are many and varied, and this data is not fit to address them all. The order data are anonymous and unlinked, so they are not suited to microfounding traders’ valuations or water management behavior. Also, the data are from one major exchange, so they open a valuable window into market operations but do not provide an account of the entire market.

RELATED LITERATURE

This paper relates to the economics literature studying formal water markets and especially its intersection with the empirical tools of industrial organization. In the southern MDB, Rafey (2023) uses a production function approach to value the water market, quantifying the increase in agricultural output caused by the existence of water trading. In mid-twentieth century Spain, Donna and Espín-Sánchez (2018, 2026) study the frictions and performance of English auctions for water among farmers. Meanwhile, where formal water markets are relatively weak or lacking, such as in California, researchers have estimated the prospective gains that could be achieved if trade barriers were eliminated (e.g., Bruno and Jessoe (2021); Hagerty (2025)). Fer-

guson (2025) studies these trade barriers, analyzing which ones are most significant, while Ferguson and Milgrom (2023) propose a new kind of water right and how to implement it. The large literature on the broader economic, legal, policy, and scientific dimensions of water markets around the globe are surveyed by Kaiser and McFarland (1997), Chong and Sunding (2006), and Debaere et al. (2014) among others. I contribute to this literature by empirically analyzing the impact of exchange design, or market microstructure, on market outcomes. I do this by exploiting not only data on realized water trades but also data on traders’ market *actions*—their buy and sell orders and order prices—some of which result in trade and some of which do not.

In structurally analyzing a limit order market, this paper relates to the pioneering work of Hollifield, Miller, and Sandås (2004) and Hollifield, Miller, Sandås, and Slive (2006) which study the Stockholm Stock Exchange and Vancouver Stock Exchange, respectively. Observably distinctive features of stock versus water markets have necessitated different modeling. For example, limit order prices for the stocks analyzed above are at most a few ticks from the best bid or ask, making for a highly discretized pricing decision, while limit order prices for water exhibit wide variation better represented as a continuous choice. Methodologically, I make new contributions to the literature by endogenizing order quantity in addition to order price (Section 4.1) and incorporating behavioral flexibility suitable for water market participants (such as risk aversion in Section 4.2). The Supplemental Appendix (A.2) additionally presents a model with dynamics from order revision and establishes identification thereof.⁵

My empirical approach is also related to work on structural estimation of auction models. Identifying and estimating players’ latent values or costs through empirical expressions of their first-order condition, rather than through numerical simulation of potentially intractable equilibria, has become a workhorse of the auction literature since Guerre, Perrigne, and Vuong (2000). A related approach has been fruitfully employed to study electricity markets (e.g., Wolak (2000); Hortagsu and Puller (2008); Reguant (2014)).

When it comes to exchange design, the finance literature has long debated the merits of continuous markets versus periodic (batch) uniform-price auctions in the context of securities exchange, such as in Madhavan (1992).⁶ Generally, there is no

⁵The model with order revision is placed in the appendix because the certainty equivalent of the continuation value ends up being small, so that revision dynamics have only a minor impact on the empirical results.

⁶Batch auctions have also been proposed more recently as a policy response to high-frequency

one-size-fits-all solution; e.g., temporal consolidation may be more helpful for thinly traded securities.

The paper is organized as follows. Section 2 describes the institutional background of water exchange in Australia, the data, and notable empirical patterns. Section 3 presents the baseline model and establishes identification of the model primitives, while Section 4 provides model extensions. Section 5 describes the estimation procedure and discusses estimation results. Section 6 presents the post-estimation analysis—including assessments of exchange performance, design, and drought interactions—and proposes a design reform. Section 7 concludes. The Supplemental Appendix (A.1) collects all proofs.

2 Water exchange

2.1 Institutional background

This section provides institutional background for water exchange in Australia.⁷ The Murray-Darling Basin is a large area of south-eastern Australia where water flows through a system of interconnected rivers and lakes. Water is traded under a cap-and-trade system as follows. The Murray-Darling Basin Agreement limits total water extraction, setting aside water for conveyance—water needed to “run the river” or deliver water without it evaporating or seeping into the riverbed—critical human water needs, and a conveyance reserve set aside for the next year.⁸ Only after setting aside these volumes is remaining water allocated to water license holders, who may trade the water. A water license or “entitlement” is a perpetual right to access an annual volume of water, subject to water availability. These entitlements were initially allocated by previous decisions of government, and new entitlements are generally not available for issue.⁹ “Entitlement trade” refers to the trade of these water licenses, which involves transferring a perpetual right. Meanwhile, “allocation trade” refers to the one-time trade of a specific quantity of water, as opposed to trading a license. This

trading. See Budish, Cramton, and Shim (2015), Wah and Wellman (2013), Farmer and Skouras (2012).

⁷The section summarizes information gathered from the Australian Productivity Commission (2003), National Water Commission (2011), Goesch, Donoghoe, and Hughes (2019), Australian Competition and Consumer Commission (2021), and Murray-Darling Basin Authority website.

⁸www.mdba.gov.au/water-management/allocations-states-mdba/water-critical-human-needs.

⁹Productivity Commission (2003).

paper focuses on *allocation trades*, the trade of water. As reported by Goesch et al. (2019), agriculture comprises around 70% of extractions for consumptive water use in Australia. This is mainly used for irrigation. As such, irrigation farmers known as “irrigators” are the main participants in Australian water markets, constituting the majority of commercial allocation trades.¹⁰ Irrigators’ values for water exhibit substantial heterogeneity, both across and within crop types (Rafey (2023)).

The southern MDB is divided into 15 trading zones, often defined as areas within which trade can freely occur because users in a zone are subject to the same jurisdiction and draw water from the same source point, such as a particular storage or water course. From the perspective of a water buyer or seller (trader), Australia’s zone-based water trading system commoditizes water within each zone. Trades within a zone, known as intra-zone trades, typically do not cause physical movement of water. The traded water exits the seller’s “allocation bank account” and enters the buyer’s “allocation bank account”. Even for trades between zones, an individual trade affects the parties’ right to access water rather than the immediate physical movement of water. This is because i) river operators generally do not release water to meet individual orders but manage bulk releases to meet forecasted aggregate demand based on historical usage data, weather, and other factors; and ii) inter-zone trade limits are “specified with a view to ensuring the future obligations to supply water users at different locations don’t change ‘too much’ ” in the aggregate.¹¹ An irrigator is free to use the water in their account at any time, though there are state rules governing the carryover of water past the end of the water year, which begins on July 1st and ends on June 30th.

Buyers and sellers are free to find a trade partner and set their trade price and volume in any manner they choose. With increased internet access, online exchanges have become one of the main avenues for trade. Participants also trade by calling brokers or may trade without intermediary involvement. Once a trade is formed, details of the transaction must be lodged on trade forms with state-owned approval or registration authorities. According to data from the Victorian Water Register, about 90% of commercial allocation trades in the state were lodged digitally through an intermediary platform,¹² and roughly 70% of these are formed online according to

¹⁰Other market participants include environmental water managers, irrigation infrastructure operators, and investors among others.

¹¹Australian Competition and Consumer Commission (2021).

¹²The application source is recorded as “broker portal” or “broker APT” for such trades.

a 2021 report.¹³

Information on trading activity, however, is highly fragmented and incomplete. There is “currently no single entity responsible for, or capable of, gathering the necessary data.”¹⁴ Importantly, my research question requires information on the *pre-trade* actions of buyers and sellers—namely, information on buy and sell orders, not all of which become trades—which is not collected by state authorities. Given the absence of order data covering the entire market, I take a focused approach, studying one major water exchange. While there are several other exchanges, the exchange I study is named by a government report as a large and significant exchange to which traders look to inform their market expectations.

As for the traded products, I study the most frequently traded allocation products so that the sample size of orders and trades is sufficient to allow estimation of exchange properties. Specifically, I study the two most traded products, water sourced from Zone 7 and from Zone 1A. The analysis of these two products serves two purposes. First, I replicate all of my Zone 7 analysis with Zone 1A data to discern which empirical findings are robust across the zones. The main text displays results for Zone 7, while the Supplemental Appendix (A.5) displays the corresponding results for Zone 1A. Second, while trade between these zones is closed for most of my data sample, there are certain periods where interzone trade is open between them; data from that combined market proves informative in Section 6.5.

2.2 The exchange

The exchange operates as follows. Both buyers and sellers, which I refer to collectively as traders, may post listings on the exchange which list the price per megaliter and quantity of water (megaliters) they wish to trade. These listings are visible to all traders. The listings do not reveal identities; the exchange operates as a blind market. If a buyer comes along and chooses one of the existing sell listings to fulfill, or a seller comes along and chooses one of the existing buy listings to fulfill, this results in a trade at the price and of the quantity listed in the listing. So traders face a choice between fulfilling an existing listing posted by another trader, for guaranteed trade, or posting their own new listing in the hope that a counterparty will come along that wants to fulfill it. I use the term “market order” to refer to the former action and the

¹³Australian Competition and Consumer Commission (2021), Table 10.3.

¹⁴Australian Competition and Consumer Commission (2021).

term “limit order” to refer to the latter action. I will refer to this market design as a “limit order market.” Conditional on trade, state authorities levy a processing fee, which was 47.50 AUD per trade during my sample period. The exchange additionally charges a fee of 55 AUD per trade and 2.2% of the trade value to both buyers and sellers. To be clear, there are no fees for placing orders; fees occur only when a trade actually occurs.

The most familiar example of a limit order market is a stock exchange, but the reader should not infer that this water exchange shares features of familiar stock exchanges other than what is described in the preceding paragraph. The water exchange is a pure limit order market with no market maker. Unlike pure limit order markets for securities, the exchange has no automatic aggregating device; e.g., a buy order is not automatically executed against the cheapest units from the aggregate supply curve. It features an open limit order book, meaning each trader can observe the set of all existing listings, and they can choose which listing to fulfill. As is typical of pure limit order markets, prices are discriminatory; different trades can happen at different prices even at the same moment in time. Meanwhile, it is optional for traders to allow their limit order quantity or “parcel” to be divided into smaller trades, and such trades do not appear prevalent in my data. A parcel divided among n counterparties would incur state and exchange fees that increase in n . This fee structure is inherited from the historically one-to-one nature of traditional water brokerage.

It is also worth noting that water is thinly traded relative to securities. According to Victorian Water Register records, the two most frequently traded allocation products—water sourced from Zones 7 and 1A—counted about 4000 and 3800 total trades through all channels, respectively, during the calendar year 2020. This is an order of magnitude less frequent than even those securities the Securities and Exchange Commission (2019) labels as thinly traded. This may magnify the impact of exchange design as design-related efficiency losses—of which some vanish as the number of participants approaches infinity—become proportionally more significant.

2.3 Data

I observe the price, volume, start time, and end time of each buy and sell limit order on the exchange, as well as the price, volume, and approval time of each trade reported by the exchange platform. The dataset covers the time period of April 2020

Table 1: Summary statistics of limit orders in the sample, Zone 7

	count	mean	sd	median	min	max
sell limit orders						
price (AUD/ML)	2,182	228	140	200	55	950
volume (ML)	2,182	61	101	30	1	1,000
duration (days)	2,182	8	12	3	0	130
buy limit orders						
price (AUD/ML)	631	140	63	118	39	407
volume (ML)	631	131	165	87	1	1,000
duration (days)	631	6	7	3	0	46

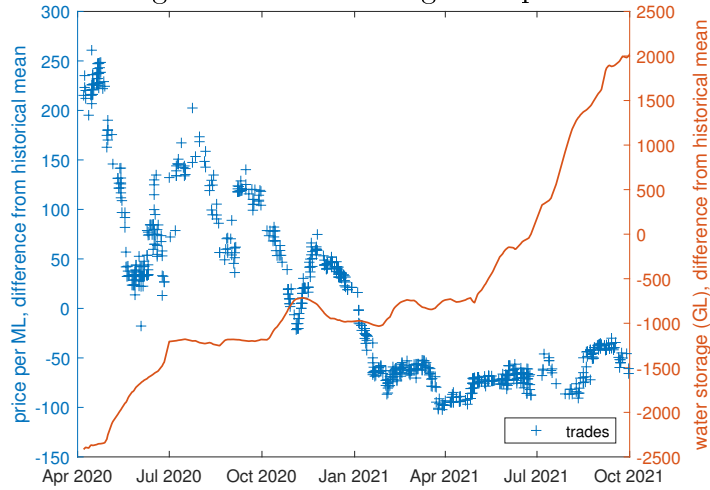
through the end of September 2021. This timeframe covers periods of both drought and relative plenty; to elaborate, water storage levels in Lake Hume, a reservoir impounded by a major dam across the River Murray, was at 13% of total capacity at the beginning of April 2020 and was at 97% of total capacity at the end of September 2021.¹⁵ This variation allows me to study the interaction of drought and exchange design in Section 6. Table 1 provides descriptive statistics of the buy and sell *limit* orders, or posted listings, for Zone 7. The mean duration between the start date and end date of a limit order is 8 days, though some limit orders have a duration as long as a few months. A large majority of limit orders, 78%, are sell limit orders, so the most representative path to trade on this exchange is for a seller to place a *limit* order (a listing) and for a buyer to fulfill it through a *market* order (purchase of a seller’s listing). To clarify, the full supply and demand on the exchange would correspond to the sum of limit orders *and* market orders; Table 1 summarizes limit orders only.

The ACCC report notes that inflows into existing public storages are the key driver of total resource availability. Indeed, when it comes to explaining water prices, the amount of water stored in major dams and lakes has a lot of explanatory power, as illustrated in Figure 1. In this figure, each cross marks a trade on the exchange. The x-axis shows the date of the trade and the left y-axis shows the deviation of the trade’s price from the historical calendar-day mean.¹⁶ The solid line, read by the right y-axis, displays the deviation of total relevant water storage from its historical

¹⁵Australia Bureau of Meteorology, <https://mdbwip.bom.gov.au/175/#4.6/-31.5/147>

¹⁶I compute the historical calendar-day mean price using all commercial allocation trades from July 1, 2009 to June 30, 2019 from the relevant zone, as reported by the Victorian Water Register. Specifically, a kernel regression of price on the day number of the water year—where July 1st is day 1—is used to predict the expected price for each month-day combination.

Figure 1: Water storage and prices



calendar-day mean.¹⁷ The respective historical calendar-day means capture expected seasonal fluctuations, while deviations from these means capture drier-than-usual or wetter-than-usual conditions for the time of year. Figure 1 shows a clear negative correlation between deviations in water storage volumes and water prices during the sample period. Section 5.2 performs regression analyses of prices with a longer list of explanatory variables.

The availability of order-level data offers a great opportunity to analyze an exchange; as I explain in later sections, the choices contained in traders’ orders are highly revealing. At the same time, the exchange data have important limitations that relate to the industry’s broader data challenges. First, trader identities are not provided, not even in anonymized form. Interestingly, the ACCC has recently recommended a new rule requiring traders to include a unique identifier on trade forms not required before. Second, the data does not provide linkages between buy and sell orders and final trades. The “lack of linkage between exchange [...] data and Basin State register data” has also been noted by the Australian government.¹⁸ Also, the set of trades reported by the platform include, without distinction, both trades from the exchange and external trades that use its form lodgment services. I will infer the linkages as well as I can by matching the price, volume, and end dates of the buy and sell orders with the price, volume, and approval dates of the trades, allowing for a reasonable delay between the two dates as state approval of lodged trades is

¹⁷I defer a detailed description of the storage variable to Section 5.2.

¹⁸Australian Competition and Consumer Commission (2021).

not always instantaneous. The inferred links may contain errors, but what they do indicate at a minimum is that a trade of matching price and volume occurred shortly after a limit order was taken off the exchange. The Supplemental Appendix (A.4) provides additional details on order-trade matching.

2.4 Empirical patterns

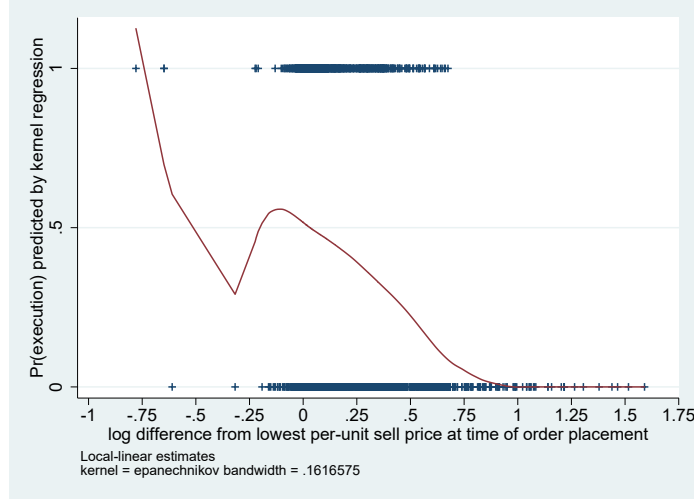
Imperfect divisibility of parcels Define aggregated exchange as an exchange in which each buy (sell) market order executes against the aggregate supply (demand) curve. This means the buyer would fill her desired quantity with the lowest-priced units (say, megaliters) available on the exchange, regardless of how many distinct sellers these units come from and without having to buy the entirety of any seller’s multi-unit parcel. In aggregated exchange, the lowest-priced units are always first to be included in a market order purchase. Pure limit order markets for securities are a familiar example.

Aggregated exchange does not characterize trade on the observed water exchange: I observe that in about 80% of buy market orders, the lowest-priced units from the relevant zone are not included in the purchase. Instead, the buyer often buys a parcel of a different quantity than the one with the lowest-priced units.¹⁹ This implies that, unlike in aggregated exchange, component units are not freely extractable from the multi-unit parcels they comprise. In Australia, the institutional fee structure disincentivizes limit orderers from permitting arbitrary or unit-level splits of the parcels they list, as each additional counterparty would be lodged as a separate trade, incurring flat fees in each instance. Effectively, the exchange then deals in parcels rather than units; the exchange matches traders to listed parcels rather than perform aggregated exchange of the component units. Parcels are heterogenous in size.

Incentives to shade limit order prices An important statistic that bears directly on traders’ pricing incentives is the probability of limit order execution as a function of limit order price. In Figure 2, the x-axis displays the log difference between the price per-ML of each sell limit order and the lowest available sell limit order price at the time that limit order was placed. The latter price is meaningful as the lowest competing price the seller would have seen before choosing a price for her own limit

¹⁹This pattern remains regardless of whether I define the lowest price in terms of pure water price or price including buyer fees.

Figure 2: Sell limit order price and execution probability, Zone 7



order. I refer to this x-axis variable as relative price. To reflect the price-per-ML that buyers would perceive, it includes buyer fees. On the y-axis, all the observed limit orders, marked by crosses, are at either 0 or 1, with 0 indicating that the limit order was not executed and 1 indicating that it was. Finally, the curve in the figure displays a kernel regression of limit order execution on relative price and can be interpreted as the predicted probability of order execution given the relative price of the order. Focusing on where data points are dense, roughly between the 1st and 99th percentiles of relative price or $x \in [-0.1, 1.1]$, the predicted $\text{Pr}(\text{execution})$ is decreasing in the seller's relative limit order price as expected; buyers prefer lower prices. What is surprising is that the slope of this decrease is rather flat; a sell limit order with a per-unit price 20% higher than the lowest available price still has a roughly 40% probability of execution. The flatter is this slope, the greater is the incentive for sellers to choose a higher limit order price because the associated loss in $\text{Pr}(\text{execution})$ is smaller. I refer to traders listing prices different from their true valuation, or seller markups and buyer markdowns, as “price shading.” Even when there is room for a Pareto-improving trade between a buyer and seller, price shading can prevent it from happening. The observed exchange exhibits strong incentives for traders to shade their limit order prices.

3 Baseline model and identification

In order to ultimately assess counterfactual exchange design, I set an intermediate goal of recovering buyers’ willingness to pay and sellers’ willingness to accept from observed data. In service of this goal, I model a trader’s pricing decision as a best response to the market she observes given her willingness to pay or accept. Let x be a vector representing current market conditions, such as water storage levels, the season, the limit order book, and recent trading activity. I refer to these collectively as “covariates” and describe them in detail in Section 5.2. Conditioning on relevant elements of x is implicit in the theory sections unless explicit conditioning is needed for clarity.

In the baseline model of Section 3, trader i ’s order quantity q_i is treated as exogenous with respect to the exchange, or as determined by factors external to the exchange. The main implication is that the same trader would bring the same quantity to a counterfactually designed exchange. Thus, the baseline model focuses on identifying each trader’s willingness to pay or accept at the *observed* q_i , which remains the relevant quantity in the counterfactual. Meanwhile, Section 4.1 develops an extended model where the trader’s order quantity is endogenous with respect to the exchange. Then trader i ’s order quantity in a counterfactually designed exchange could differ from the observed q_i , and counterfactual analysis requires identification of traders’ values at quantities $\neq$ observed q_i . In further extensions of the model, Section 4.2 accommodates risk aversion, and the Supplemental Appendix shows how to accommodate dynamics from order revision.

3.1 Limit order pricing

Buyers and sellers (traders) arrive exogenously to a market for water. I initially focus on buyers, returning to sellers in Section 3.3. For order quantity q , buyer i obtains value v_i per unit—implicitly $v_i(q)$ —upon obtaining the water.

Upon arrival, buyer i either places a buy listing at price p_i or fulfills an existing sell listing. Fulfilling a sell listing involves buying from the seller who placed the listing at the terms stated in the listing. In this paper, I refer to placing one’s own listing as a “limit order” and fulfilling an existing listing as a “market order”. While a market order always results in a trade, a limit order becomes a trade only if a counterparty fulfills it. Thus, limit order execution is not guaranteed. I refer to the set of all live

buy and sell limit orders not yet executed as the “limit order book”.

Let $P(p, q)$ denote the probability that a buy limit order with price p and quantity q is executed. This probability, an empirically observed object, is an input to the buyer’s best-response problem. I make the following assumptions regarding $P(p, q)$.

Assumption 1. *The execution probability $P(\cdot, \cdot)$ for a buy limit order satisfies*

- (i) *Monotonicity:* $P(p, q)$ is strictly increasing in p .
- (ii) *Decreasing reversed hazard rate:* $P(p, q)/P_1(p, q)$ is strictly increasing in p .
- (iii) *Uncertain execution:* $P(p, q) < 1$.

Assumption 1-(i) states that a buy limit order with a higher price is more likely to be executed. This is because the counterparties—sellers—prefer to receive higher prices. Assumption 1-(ii) is a technical condition involving the derivative $P_1(p, q) \equiv \frac{\partial P(p, q)}{\partial p}$. Assumption 1-(iii) states that execution of a buy limit order is not guaranteed.²⁰

Now consider the buyer’s choice of limit order price. I adopt a best-response approach, in the spirit of Guerre, Perrigne, and Vuong (2000) and Hollifield, Miller, and Sandås (2004). Given her value v_i and quantity q_i , buyer i chooses price p to maximize her expected utility from the order in light of execution probability function $P(\cdot, \cdot)$. Her maximization problem is

$$\max_p P(p, q_i)[(v_i - \tau p)q_i - f], \quad (1)$$

where her objective function is the execution probability times her profit conditional on execution. Trade fees levied as a percentage of the price are included in $\tau > 1$, while f is a flat fee per trade. Differentiating the objective function with respect to p , the first-order condition gives

$$v_i - \left(\tau p + \frac{f}{q_i} \right) = \tau \frac{P(p, q_i)}{P_1(p, q_i)} > 0. \quad (2)$$

The expression $\tau p + f/q_i$ is the total price per unit including fees. The left-hand side of (2) thus represents the buyer’s markdown; rather than listing a price at which she breaks even, the buyer marks down or shades her limit order price so that the fee-inclusive price is less than her true value: $\tau p + f/q_i < v_i$. The right-hand side

²⁰Cohen et al. (1981) show that unless traders monitor a market continuously, the execution probability of a limit order is less than one no matter how closely it approaches the asking price.

shows that the markdown is larger when $P_1(p, q_i)$ is smaller, meaning when the slope of execution probability with respect to price is less steep.

Let $p^*(v_i, q_i)$ denote the buyer's optimal limit order price that satisfies equation (2). Under Assumption 1, $p^*(v_i, q_i)$ is strictly increasing in v_i , so ceteris paribus, buyers with higher values for water post higher limit order prices. Equation (2) also shows that limit order pricing relies on dividing the flat fee f over q_i units. Thus, listing a price of p^* does not imply that the buyer is willing to buy arbitrary quantities at that price. For example, purchasing just one unit at price p^* , or one unit each from several different sellers, would incur an expense of $\tau p^* + f$ per unit, which could exceed v_i and yield a negative payoff even if $\tau p^* + f/q_i < v_i$. It is understandable that traders would opt out of arbitrary parcel splits when they set their limit orders.

The maximization problem in (1) does not require that the payoff of non-execution be zero. For example, if the payoff of no execution is V_0 , the buyer's pricing decision becomes $\max_p P(p, q)[(v' - \tau p)q - f] + (1 - P(p, q))V_0$, which has the same solution as $\max_p P(p, q)[(v' - v_0 - \tau p)q - f]$, where $v_0 \equiv V_0/q$. Compared to (1), the only change is that $v' - v_0$ takes the place of v . Thus, if $V_0 \neq 0$, we can interpret v as $\equiv v' - v_0$, the value-added of obtaining water here versus not obtaining it.

3.2 Choosing between a market order and a limit order

The previous section considered the buyer's choice of what price to post given that she places a limit order. Now consider the buyer's choice between placing a market order and a limit order. A rational buyer would compare the profit of the market order to that of a limit order and choose the one whose expected profit is higher.

Suppose a buyer places a market order resulting in a purchase of q units at per-unit price ℓ . Let the buyer obtain value v per unit from this purchase. Execution of market orders is guaranteed by definition, so the buyer's per-unit profit from this market order is simply

$$M(v, \ell) \equiv v - \left(\tau \ell + \frac{f}{q} \right). \quad (3)$$

For the buyer to have placed this market order, it must be that $M(v, \ell)$ exceeds the expected profit from placing a limit order. The buyer might obtain a lower price with a limit order, but the tradeoff is that limit order execution is uncertain. The buyer's

expected per-unit profit from placing an optimally priced limit order is

$$L(v) \equiv \left[v - \left(\tau p^*(v, q) + \frac{f}{q} \right) \right] P(p^*(v, q), q). \quad (4)$$

The buyer placing a market order implies that her v satisfies $M(v, \ell) \geq L(v)$. The next proposition states that the buyer's preference between $M(v, \ell)$ and $L(v)$ is delineated by a threshold such that buyers with values lower than the threshold, denoted $\tilde{v}(\ell)$, would prefer $L(v)$ and buyers with values higher than the threshold would prefer $M(v, \ell)$. Intuitively, the guarantee of trade offered by market orders is more valuable for higher-value buyers.

Proposition 1. *Among $v > \tau\ell + f/q$, there is at most one value such that $L(v) = M(v, \ell)$, i.e., such that a buyer with this value is indifferent between an optimally priced limit order and a market order of price ℓ . Denoting that value as $\tilde{v}(\ell)$, the buyer would prefer $L(v)$ if $v < \tilde{v}(\ell)$ and prefer $M(v, \ell)$ if $v \geq \tilde{v}(\ell)$.*

Meanwhile, data suggest that some traders arriving in the market might default to placing a limit order, rather than placing a limit order only if it is more profitable than a market order. For example, there is a market norm of sellers listing their parcels for sale (placing limit orders). I allow for this default behavior in buyers and sellers with probability α_b and α_s , respectively. With the remaining probability, I let traders act in accordance with Proposition 1.

3.3 Seller analogs

Now consider the seller. For order quantity q , let seller i 's opportunity cost of relinquishing the water be c_i per unit—implicitly $c_i(q)$. I now adopt b and s subscripts to distinguish notation for the buyer and seller, respectively. The seller's problem is largely a mirror image of the buyer's. Therefore, this section concisely summarizes seller analogs of the results discussed above for buyers. Note that unlike buy limit orders, sell limit orders with a higher price are *less* likely to be executed because the counterparties—buyers—prefer to pay lower prices. Thus, the seller analog of Assumption 1 (“Assumption 1S”) replaces “increasing” with “decreasing” in part (i).

Limit-order pricing Consider the seller's choice of limit order price when placing a limit order. Given her opportunity cost c_i , quantity q_i , and probability of order

execution $P_s(\cdot, \cdot)$, her maximization problem is $\max_p P_s(p, q_i)[(\tau_s p - c_i)q_i - f_s]$. Trade fees levied as a percentage of the price are represented by $\tau_s < 1$, while f_s is a flat fee per trade. The associated first-order condition for limit-order pricing is

$$c_i - \left(\tau_s p - \frac{f_s}{q_i} \right) = \tau_s \frac{P_s(p, q_i)}{P_{s,1}(p, q_i)} < 0. \quad (5)$$

The expression $\tau_s p - f_s/q_i$ is the price per unit the seller actually receives net of all fees, and $P_{s,1}(p, q_i)$ is the derivative of execution probability with respect to price. The right-hand side is negative because $P_{s,1}(p, q_i) < 0$. Thus, rather than posting a price at which she breaks even, the seller marks up her price such that $\tau_s p - f_s/q_i > c_i$. Let $p_s^*(c_i, q_i)$ denote the optimal limit order price that satisfies equation (5). Given Assumption 1S, $p_s^*(c_i, q_i)$ is strictly increasing in c_i , so sellers with higher opportunity costs post higher limit order prices ceteris paribus.

Choosing between a market order and a limit order Suppose a seller places a market order resulting in a sale of q units at per-unit price h . Let the seller's opportunity cost of this sale be c per unit. Then the seller's per-unit profit from this market order is $M(c, h) \equiv (\tau_s h - f_s/q) - c$. For the seller to have placed this market order, it must be that $M(c, h)$ exceeds the expected profit from placing a limit order. The seller's expected per-unit profit from placing an optimally priced limit order is $L(c) \equiv [(\tau_s p_s^*(c, q) - f_s/q) - c] P_s(p_s^*(c, q), q)$. A loose summary of Proposition 1S is that low-cost sellers prefer market orders, while high-cost sellers prefer limit orders.

Proposition 1S. *Among $c < \tau_s h - f_s/q$, there is at most one value of c such that $L(c) = M(c, h)$, i.e., such that a seller with this cost is indifferent between an optimally priced limit order and a market order of price h . Denoting that value as $\tilde{c}(h)$, the seller would prefer $L(c)$ if $c > \tilde{c}(h)$ and prefer $M(c, h)$ if $c \leq \tilde{c}(h)$.*

3.4 Identification of the baseline model

I assume the following are observed for each limit order: the price p , quantity q , and covariates x ; and indicators for whether it is buy or sell and executed or not. The following are observed for each market order: the resulting trade price p , quantity q , covariates x , and whether it is buy or sell. For the purpose of assessing counterfactual exchange design, the necessary model objects to identify are buyers' values v and

sellers' costs c . This section explains how these are point-identified at the *order-level*, or observation by observation, for all limit orders. For market orders, we obtain order-level bounds on each v and c . In keeping with convention and for the sake of expositional brevity, I use distributional notation to concisely refer to the collection of all this information: $F_b(v|q, x)$ and $F_s(c|q, x)$, or the conditional “distributions” of buyers' values and sellers' costs. However, the notation should not confuse the fact that more granular identification is in fact achieved—and subsequently exploited in counterfactual analysis. As intermediate steps, I also identify the probabilities of limit order execution, $P_b(p, q|x)$ and $P_s(p, q|x)$, and the probabilities of defaulting to limit orders, α_b and α_s . For the remainder of this section, I abstract away from conditioning on x .

$P_b(\cdot, \cdot)$ and $P_s(\cdot, \cdot)$ The probability of limit order execution $P_b(p, q)$ and $P_s(p, q)$ for the buyer and seller, respectively, are identified as $E[\mathbb{1}_{\text{executed}}|p, q]$, where $\mathbb{1}_{\text{executed}} = 1$ if a limit order is executed and 0 otherwise.

v_i or c_i of each limit order Having identified $P_b(\cdot, \cdot)$ and $P_s(\cdot, \cdot)$, the buyer's v_i or seller's c_i associated with each limit order is identified by rearranging first-order conditions (2) and (5) as

$$v_i = \left(\tau_b p_i + \frac{f_b}{q_i} \right) + \tau_b \frac{P_b(p_i, q_i)}{P_{b,1}(p_i, q_i)}, \quad (6)$$

$$c_i = \left(\tau_s p_i - \frac{f_s}{q_i} \right) + \tau_s \frac{P_s(p_i, q_i)}{P_{s,1}(p_i, q_i)}. \quad (7)$$

Given observed p_i, q_i and the fee structure τ_b, f_b, τ_s, f_s , the right-hand sides of (6) and (7) can be computed directly to identify v_i and c_i . The inverse is that these equations characterize the optimal pricing function $p^*(\cdot, \cdot)$; that is, given any v or c , we can (numerically) solve for the optimal limit order price.

α_b and α_s Per Proposition 1, a rational buyer i prefers a market order of price ℓ_i over a limit order if $v_i \geq \tilde{v}(\ell_i)$, where the threshold $\tilde{v}(\ell_i)$ is defined as the unique solution to the indifference condition $L(\tilde{v}(\ell_i)) = M(\tilde{v}(\ell_i), \ell_i)$. We can solve for $\tilde{v}(\ell_i)$ because every component of this equation is either observed, known, or identified above; see (3) and (4) for definitions of $L(\cdot)$ and $M(\cdot, \cdot)$. The probability α_b that

buyers default to placing a limit order is then identified as the probability that they place a limit order despite $v_i \geq \tilde{v}(\ell_i)$:

$$\alpha_b = \frac{\#\{\text{limit orders} | v_i \geq \tilde{v}(\ell_i)\}}{\#\{\text{market orders}\} + \#\{\text{limit orders} | v_i \geq \tilde{v}(\ell_i)\}}. \quad (8)$$

Similarly, by Proposition 1S, the probability α_s that sellers default to placing a limit order is identified as the probability that they place a limit order despite $c_i \leq \tilde{c}(h_i)$:

$$\alpha_s = \frac{\#\{\text{limit orders} | c_i \leq \tilde{c}(h_i)\}}{\#\{\text{market orders}\} + \#\{\text{limit orders} | c_i \leq \tilde{c}(h_i)\}}. \quad (9)$$

$F_b(\cdot|\cdot)$ and $F_s(\cdot|\cdot)$ While the buyer's v associated with each limit order is point identified as explained above, all we can know about the v associated with each observed market order of price ℓ_i is a lower bound, that $v_i \geq \tilde{v}(\ell_i)$. Thus, for fixed ℓ , we would not be able to identify the value distribution $F_b(\cdot|\cdot)$ above $\tilde{v}(\ell)$ if $\alpha_b = 0$; we would only be able to identify the shape of $F_b(\cdot|\cdot)$ for values up to $\tilde{v}(\ell)$ and identify the total mass but not the shape of values above $\tilde{v}(\ell)$. This could be sufficient for parametric identification of $F_b(\cdot|\cdot)$ but not for nonparametric identification. On the other hand, if $\alpha_b > 0$, we would also see limit orders associated with values above $\tilde{v}(\ell)$. Since we can identify the v_i associated with each limit order, this allows for the nonparametric identification of $F_b(\cdot|\cdot)$ on its entire support. Analogously, if $\alpha_s > 0$, $F_s(\cdot|\cdot)$ is nonparametrically identified on its entire support. The proof of Proposition 2 provides the details of this step.

Proposition 2. *Under Assumptions 1 and 1S, the buyer value v or seller cost c associated with each limit order is identified in addition to the probabilities of defaulting to limit orders, α_b and α_s . If $\alpha_b, \alpha_s > 0$, the value and cost distributions $F_b(\cdot|\cdot)$ and $F_s(\cdot|\cdot)$ are also identified nonparametrically.*

4 Model extensions

4.1 Endogenous order quantity

This section extends the baseline model of Section 3 to the case of order quantities that are endogenous with respect to the exchange. As before, let buyer i obtain value $v_i(q)$ per unit upon obtaining q units of water; to be clear, this is an average value

across the q units in the parcel. Unlike before, quantity q_i is no longer fixed for i . It now becomes explicit that buyer i 's value for water is described by a function $v_i(\cdot)$. This function is specific to buyer i , takes q as an argument, and determines buyer i 's local demand curve. The goal of this section is to establish what we can learn about $v_i(\cdot)$ from the data given a model of endogenous q , with an eye to enabling analysis of counterfactual exchange design.

Continuous quantity choice When placing a limit order, the buyer now optimizes both price and quantity with respect to the exchange, according to

$$\max_{p,q} P(p,q)[(v_i(q) - \tau p)q - f].$$

Subscripts b to indicate buyers, such as in τ_b or f_b , are omitted for brevity. The solution (p,q) to this optimization problem corresponds to the solution to the following system of equations.

$$v_i(q) = \tau p + \frac{f}{q} + \tau \frac{P(p,q)}{P_1(p,q)}, \quad (10)$$

$$v'_i(q) = \left(\tau p - v_i(q) - \frac{P_2(p,q)}{P(p,q)}[(v_i(q) - \tau p)q - f] \right) / q. \quad (11)$$

Equation (10) is identical to equation (2) and comes from the optimality of p given q , characterizing $p^*(q)$. Equation (11) comes from optimality of q given $p = p^*(q)$ and the envelope theorem.

Now consider identification, or what we can learn about the function $v_i(\cdot)$, assuming the observed (p_i, q_i) satisfy this system of equations. Equation (10) identifies $v_i(q_i)$ at the observed q_i , as before. But now, the buyer's observed choice of q_i gives us additional information about the buyer's demand *curve*. Specifically, we can compute the right-hand side of (11) from observables and knowledge of $v_i(q_i)$, so (11) identifies $v'_i(q_i)$. This is the local slope of $v_i(\cdot)$, or how buyer i 's average value per unit changes with quantity, local to the q_i she orders in the data. This is useful, especially if we expect counterfactual quantities to change but not change drastically.

Meanwhile, if we are willing to impose a parametric specification on $v_i(\cdot)$, then knowledge of $v_i(q)$ and $v'_i(q)$ is sufficient to identify a scale and curvature parameter for each i . Specifically, let $V_i(q) \equiv v_i(q)q$ denote a buyer's total value from obtaining

q units. We can specify

$$V_i(q) = a_i q^{b_i}, \quad (12)$$

where $a_i \geq 0$ is a scale parameter, $b_i > 0$ is a curvature parameter, and $V_i(0) = 0$. In particular, b_i governs how buyer i 's value changes with quantity; e.g., $b_i \in (0, 1)$ would indicate decreasing marginal values. With knowledge of $v_i(q_i)$ and $v'_i(q_i)$ given by (10)-(11), the curvature parameter is identified as $b_i = 1 + q_i v'_i(q_i)/v_i(q_i)$, and subsequently the scale parameter is identified as $a_i = V_i(q_i)/q_i^{b_i}$. I assume that, where this model applies, buyers have weakly decreasing marginal values and sellers have weakly increasing marginal costs, in keeping with intuition.

Discrete quantity choice When quantity choice is continuous, the scale and curvature parameters of $V_i(q) = a_i q^{b_i}$ are point identified, as explained above. When quantity choice is discrete, we obtain bounds on the parameters instead of point identification. I will illustrate these bounds for the case of market orders, where the set of parcels on offer by the other side is likely finite.

Suppose we observe a buyer make a market order purchase described by (p_i, q_i) . Suppose other parcels of (p', q') were also available for purchase but foregone in favor of (p_i, q_i) . By revealed preference, this implies the inequality $V_i(q') - \tau p' q' < V_i(q_i) - \tau p_i q_i$. For any given value of $V_i(q_i)$ at purchased q_i , rearranging the inequality yields the following bounds on the curvature parameter b_i .

$$b_i < \min_{q' \in \{q' | q' > q_i\}} \frac{\ln(V_i(q_i) - \tau p_i q_i + \tau p' q') - \ln V_i(q_i)}{\ln q' - \ln q_i}, \quad (13)$$

$$b_i > \max_{q' \in \{q' | q' < q_i\}} \frac{\ln(V_i(q_i) - \tau p_i q_i + \tau p' q') - \ln V_i(q_i)}{\ln q' - \ln q_i}. \quad (14)$$

Intuitively, foregoing a larger parcel implies the foregone units are worth less than the cost difference between the parcels, while foregoing a smaller parcel implies the added units are worth more than the cost difference between the parcels. These lead to upper and lower bounds, respectively, on parameter b_i , as specified in (13)-(14). The richer the set of alternative parcels that were available, the tighter the bounds become.

Recall that $V_i(q_i)$ at observed q_i is point-identified for each limit order, while each market order implies a bound on $V_i(q_i)$. Therefore, for each market order, we now obtain (joint) bounds in the space of $(V_i(q_i), b_i)$, equivalent to bounds in (a_i, b_i) -space,

since $a_i = V_i(q_i)/q_i^{b_i}$.

Discussion We have established a model of endogenous q and what it identifies about values at counterfactual quantities. This is valuable because it allows us to check that our conclusions are robust to endogenously changing q . However, some traders may not be able to adjust their quantity in practice. For example, a seller may have no more water to sell than what she has posted on the observed exchange. Allowing such sellers to supply more units in the counterfactual could artificially inflate the benefits of exchange reform. Similarly, a buyer may have lumpy demand, with little use for water below some minimum amount. As such, counterfactual analysis should permit data-driven model exceptions and err on the side of conservative welfare gains.

4.2 Risk aversion

The main participants in Australian water markets are irrigators (farmers). In light of evidence from the literature, e.g., Bajari and Hortaçsu (2005), that real behavior is often better represented by risk-averse models than risk-neutral ones, I allow for the possibility that traders are risk averse. A risk-averse buyer would choose a limit order price to maximize $P(p_i, q_i)U((v_i - \tau p_i)q_i - f)$, where $U(\cdot)$ is the utility function. Under the constant relative risk aversion (CRRA) specification of $U(\cdot) = \frac{(\cdot)^{1-\eta}}{1-\eta}$, the first-order condition for this maximization problem becomes

$$v_i - \left(\tau p_i + \frac{f}{q_i} \right) = (1 - \eta) \tau \frac{P(p_i, q_i)}{P_1(p_i, q_i)}. \quad (15)$$

This is the risk averse analog of equation (2). A CRRA parameter of $\eta = 0$ represents risk neutrality while $\eta > 0$ represents risk aversion. Given Assumption 1, a more risk-averse buyer will choose a higher limit order price to satisfy (15); i.e., she will mark down (shade) her price less. The intuition is that a risk-averse trader is less tolerant of the risk of not trading and willing to accept lower profit to reduce this risk. Importantly, for any given η , the v_i associated with an observed limit order price p_i is identified by rearranging equation (15) to solve for v_i .

When it comes to market orders, risk aversion lowers the threshold value above which buyers prefer a market order over a limit order. This is because the guarantee of trade offered by market orders is more attractive when traders are risk averse.

It remains to identify the risk aversion parameter η . The empirical literature shows that risk aversion can be identified, for example, through various exclusion restrictions. Here, I will take a more flexible approach: rather than pinning η down through an exclusion restriction, the empirical analysis will be conducted over a range of η values that can rationalize the data under Assumption 2 below. Per equation (15), an important role of η is to affect the amount of price shading, or the difference between the trader’s chosen limit order price and her underlying value. Conducting the analysis across a range of risk aversion levels will help us draw conclusions that are robust to a wide range of price shading behavior.

Assumption 2. *Buyer values v and seller opportunity costs c are non-negative.*

Some values of η may not be able to rationalize the data while satisfying Assumption 2. For example, consider the seller analog of equation (15), $c_i = (\tau_s p_i - f_s/q_i) + (1 - \eta)\tau_s P_s(p_i, q_i)/P_{s,1}(p_i, q_i)$. Because $P_{s,1}(p_i, q_i) < 0$ for the seller, the right-hand side of this equation could be negative at some values of η , which would violate Assumption 2.

5 Estimation

5.1 Estimation

While the estimation procedure closely aligns with the identification argument, a key task of estimation is to condition estimated functions on the vector of covariates x in the context of a limited sample. Therefore, some functions will be estimated with parametric specifications though they are identified nonparametrically. To allow a range of different risk aversion levels, I conduct the estimation procedure once for each value of the CRRA parameter η over a grid in $[0,1)$ at intervals of 0.01. This section uses the buyer model, denoted with a b subscript, to explain the estimation procedure; the seller analogs, with an s subscript, are estimated in the same manner using the seller’s version of the model equations and definitions.

Estimation of $P_b(p, q|x)$ The probability of limit order execution $P_b(p, q|x)$ is not a model primitive; it is an empirical object of which the estimation is an intermediate step for estimating buyers’ values. To condition on covariate vector x , which has

many elements, given a finite data sample, I use a probit specification of the form $P_b(p, q|x) = \Phi(f_p(p; \beta_p) + f_q(q; \beta_q) + x\beta_x)$, where $\Phi(\cdot)$ is the standard normal CDF.²¹ The functions $f_p(p; \beta_p)$ and $f_q(q; \beta_q)$ are polynomials to allow flexible shapes in p and q , and $\beta_p, \beta_q, \beta_x$ are probit coefficient vectors to be estimated. In particular, I constrain derivatives of the price polynomial $f_p(\cdot; \beta_p)$ to ensure Assumption 1, and the degree of the polynomial is determined by the Akaike Information Criterion (AIC).

Careful details go into estimating $P_b(\cdot, \cdot|\cdot)$ in the presence of fees and price distribution heterogeneity across time. With fees, the price a seller actually receives is $\tilde{p} \equiv \tau_s p - f_s/q$ per unit; this matters for $P_b(\cdot, \cdot|\cdot)$ because execution of a buy limit order depends on its attractiveness to sellers. So $f_p(\cdot; \beta_p)$ includes an adjustment from p to $\tilde{p}$. To account for potential changes of the function $P(\cdot, \cdot|\cdot)$ across time, I divide the sample into a number of time periods I define in Section 6.4 and estimate $P_s(\cdot, \cdot|\cdot)$ separately for each period-zone combination.²² In addition, as order execution depends on how competitive a limit order price is relative to the contemporary price distribution, $f_p(\cdot; \beta_p)$ normalizes the price input as (price – best available price at time of order placement)/(standard deviation of traded prices during the preceding week, smoothed) to better estimate $P_b(\cdot, \cdot|\cdot)$ given a data sample where price distributions do not stay fixed.²³

Estimation of the $\hat{v}$ associated with each limit order Having estimated $P_b(p, q|x)$, the buyer's value v_i associated with each limit order is estimated by the first-order condition (15), of which the empirical analog is

$$\hat{v}_i = \left(\tau_b p_i + \frac{f_b}{q_i} \right) + (1 - \eta) \tau_b \frac{\hat{P}_b(p_i, q_i|x_i)}{\hat{P}_{b,1}(p_i, q_i|x_i)}.$$

Estimation of α_b Having estimated the $\hat{v}_i$ of each limit order, the probability α_b that buyers default to placing a limit order is estimated using the empirical analog of

²¹The covariates are described in Section 5.2. From that list, I condition $P(p, q|x)$ on all of x_e and the covariates in Table 2, column (2).

²²To be clear, this is for sellers' $P_s(\cdot, \cdot|\cdot)$. For buyers' $P_b(\cdot, \cdot|\cdot)$, estimation pools more periods because the sample size of buy limit orders is smaller.

²³When the number of trades during the preceding week is less than 20, I use the standard deviation of the last 20 trade prices instead.

equation (8), namely

$$\hat{\alpha}_b = \frac{\#\{\text{limit orders} | \hat{v}_i \geq \hat{v}(\ell_i)\}}{\#\{\text{market orders}\} + \#\{\text{limit orders} | \hat{v}_i \geq \hat{v}(\ell_i)\}}.$$

The market/limit threshold $\hat{v}(\ell_i)$ applicable to each limit order i is found by numerically solving the indifference condition that a buyer with value $\hat{v}(\ell_i)$ is indifferent between a limit order and a market order, given the risk aversion level η .

Estimation of $F_b(v|q, x)$ Recall that v_i is not point-identified for observed market orders; rather, we have a bound $v_i \geq \hat{v}(\ell_i)$. Therefore, though I have order-level $\hat{v}_i$ for all limit orders, which I use directly, I also estimate a distribution $F_b(v|q, x)$ to simulate values specifically for those who placed market orders. Before specifying any functional form for $F_b(v|q, x)$, I find that the empirical distribution of $\hat{v}$ obtained according to the steps above approximates a lognormal distribution. (See Section 5.3 and Figure A1.) Therefore, I specify ex post a lognormal distribution for $F_b(v|q, x)$. Specifically, I let v be distributed according to $\text{Lognormal}(x\theta_\mu, \exp(x\theta_\sigma))$, where $x\theta_\mu$ and $\exp(x\theta_\sigma)$ are the mean and standard deviation of $\ln(v)$, respectively. The parameter vectors θ_μ and θ_σ are estimated by maximizing the likelihood of observed limit and market orders. The likelihood of a market order is $(1 - F_b(\hat{v}(\ell_i)|q_i, x_i))(1 - \hat{\alpha}_b)$ because a market order implies $v_i \geq \hat{v}(\ell_i)$ and that the buyer did not default to a limit order. Letting $f_b(v|q, x)$ denote the pdf of $F_b(v|q, x)$, the likelihood contribution of a limit order with $\hat{v}_i \geq \hat{v}(\ell_i)$ is $f_b(\hat{v}_i|q_i, x_i)\hat{\alpha}_b$, and the likelihood contribution of a limit order with $\hat{v}_i < \hat{v}(\ell_i)$ is $f_b(\hat{v}_i|q_i, x_i)$.

5.2 Covariates

The purpose of covariates is to account for and absorb within-sample heterogeneity in market conditions when estimating the model with a finite data sample where market conditions do not remain fixed. They broadly fall into two categories. The first category, which I collect in the vector x_w , concerns the fundamentals of water. In x_w , I represent seasonality using Bernstein polynomials of *waterday*, which is a day number: the first day of the water year, July 1st, is day 1, and the last day of the water year, June 30th, is day 365 or 366 depending on the year. I account for water availability using the daily total amount of water in the major storages relevant to

each zone, as provided by the Australia Bureau of Meteorology.²⁴ To capture true deviations of storage levels from long-term norms as opposed to expected seasonal fluctuations, I kernel regress the daily total storage amounts during the ten full water years that precede my data sample, July 1, 2009 to June 30, 2019, on *waterday* and then refer to log deviations from this historical *waterday* average as “seasonally adjusted” log storage. I also use rainfall data from the Bureau of Meteorology to compute average daily rainfall across the weather stations relevant to each zone.

The second category of covariates, which I collect in the vector x_e , concerns conditions of the exchange at the time the trader’s action being analyzed took place. While the exchange is a high dimensional object, I include a feasibly limited set of variables: the total volume in megaliters of live sell limit orders (ask depth) and buy limit orders (bid depth); the difference between the lowest sell limit order price and the highest buy limit order price (bid-ask spread); total trade volume and the mean and standard deviation of trade prices during the preceding week. If there were no trades during the preceding week, I substitute with the mean and standard deviation of trade prices from the closest preceding week in which trades occurred.

Table 2 shows that these covariates are able to explain a large fraction of the sample variance of limit order prices. The regressions in columns (1)-(3) pool buy and sell limit orders, while columns (4) and (5) use only sell limit orders and buy limit orders, respectively. The $R^2 = 0.83$ of column (1) shows that elements of x_w , the water fundamentals, have high explanatory power when it comes to limit order prices. Meanwhile, columns (2) and (3) show that the mean trade price of the preceding week largely captures the combined effects of the slow-moving components of x_w (storage level and seasonality). Columns (4) and (5) include the remaining x_e elements as regressors. In general, these regression coefficients need not have causal interpretations; the goal here is not to estimate causal effects but to absorb within-sample heterogeneity in market conditions.

5.3 Estimation results

As discussed in Section 4.2, I conduct the estimation procedure once for each value of the CRRA parameter η over a grid in $[0,1)$ at intervals of 0.01. To respect Assump-

²⁴For Zone 7, I include major storages on the Victorian Murray: Lake Dartmouth, Lake Hume, the Menindee Lakes, and Lake Victoria. For Zone 1A, I include major storages on the Goulburn: Lake Eildon and Waranga Basin.

Table 2: Regression of log limit order price on covariates

	(1)	(2)	(3)	(4)	(5)
daily storage (log), seasonally adjusted	-2.644 (0.025)		-0.264 (0.063)	-0.231 (0.086)	-0.203 (0.116)
waterday polynomial term 1	0.759 (0.053)		-0.002 (0.046)	0.020 (0.064)	0.210 (0.103)
waterday polynomial term 2	-2.052 (0.038)		-0.241 (0.055)	-0.058 (0.071)	-0.198 (0.115)
waterday polynomial term 3	-0.549 (0.025)		-0.093 (0.023)	-0.052 (0.028)	0.018 (0.038)
daily rainfall (1000mm)	-0.297 (2.901)	-5.682 (2.298)	-5.310 (2.325)	-4.417 (2.536)	1.447 (4.644)
mean trade price, prev week (log)		0.976 (0.007)	0.884 (0.022)	0.944 (0.030)	0.832 (0.047)
st dev of trade price, prev week				0.004 (0.001)	0.000 (0.001)
total trade volume, prev week (1000 ML)				-4.888 (2.276)	-3.240 (3.018)
ask depth (1000 ML)				-34.574 (4.567)	-1.188 (8.598)
bid depth (1000 ML)				45.260 (7.607)	39.010 (9.163)
bid-ask spread (100 AUD)				4.725 (1.154)	-5.516 (2.486)
log quantity (ML)				0.002 (0.003)	0.024 (0.006)
sell dummy	0.172 (0.011)	0.148 (0.009)	0.154 (0.009)		
constant	5.083 (0.018)	-0.005 (0.035)	0.506 (0.116)	0.282 (0.150)	0.552 (0.234)
Data sample: sell or buy limit orders	both	both	both	sell	buy
Observations	2813	2813	2813	2182	631
R^2	0.829	0.890	0.891	0.897	0.862
Adjusted R^2	0.829	0.889	0.890	0.896	0.859

Standard errors in parentheses

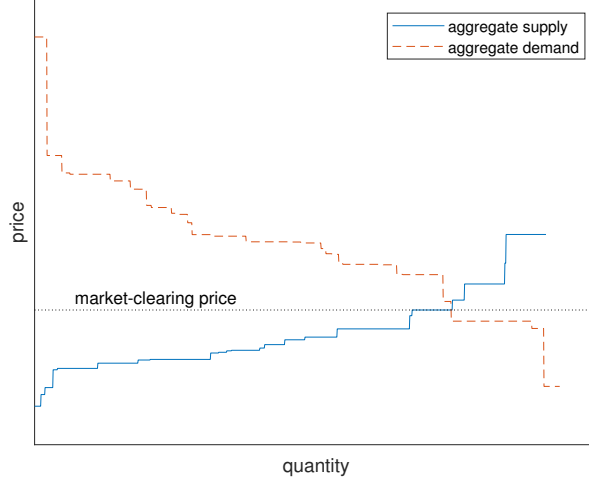
tion 2 while allowing for econometric error, the post-estimation analysis of Section 6 focuses on the range of η values given which at most 1% of all sell limit orders have $\hat{c} < 0$. This narrows η to $[0.57, 1)$. The remainder of the paper will report results at the two extremes of this range, $\eta = 0.57$ and $\eta = 0.99$.

Order-level $\hat{v}_i$ or $\hat{c}_i$ are estimated for all limit orders. To describe these order-level value estimates, it is helpful to relate them to the respective observed order prices. As discussed previously, different values of η lead to different amounts of price shading. Given $\eta = 0.99$, the median markdown $\left| \frac{p_i - (\hat{v}_i - f_b/q_i)/\tau_b}{(\hat{v}_i - f_b/q_i)/\tau_b} \right|$ among buy limit orders and the median markup $\frac{p_i - (\hat{c}_i + f_s/q_i)/\tau_s}{(\hat{c}_i + f_s/q_i)/\tau_s}$ among sell limit orders are both about 0.4% in the sample. In other words, there is very little price shading when traders are this risk averse, resulting in $p \approx v$. Meanwhile, given lower risk aversion at the level of $\eta = 0.57$, the analogous medians are 14% for buy limit orders and 19% for sell limit orders in Zone 7 (16% and 17% in Zone 1A).

The estimated probability α of placing a limit order by default varies with η . This is because buyers' $\hat{v}$ and sellers' $\hat{c}$ also vary with η , and α is estimated by assessing the rationality of choosing a limit order versus a market order in light of these $\hat{v}$ or $\hat{c}$. Nonetheless, estimates given both high and low η indicate that sellers are much more likely to default to limit orders than are buyers, in harmony with the empirical observation that there are 3-4 times as many sell limit orders as buy limit orders (see Table 1). Specifically, $\hat{\alpha}_s = 0.71$ and $\hat{\alpha}_b = 0.15$ given $\eta = 0.57$, while $\hat{\alpha}_s = 0.23$ and $\hat{\alpha}_b = 0.03$ given $\eta = 0.99$.

Finally, there are the value distribution $F_b(v|q, x)$ and cost distribution $F_s(c|q, x)$, which are used only to simulate values for traders who placed market orders; Section 6.1 describes simulation details. These distributions are conditioned on water fundamentals x_w but not on x_e , as $F_b(v|q, x_w)$ and $F_s(c|q, x_w)$, because traders' latent values and costs are deep parameters here. This is in contrast to chosen actions, like choice of limit order price given one's value, which do respond strategically to exchange conditions x_e . Echoing Table 2 column (1), $\hat{F}_b(v|q, x_w)$ and $\hat{F}_s(c|q, x_w)$ reflect water values that are decreasing in the amount of water available in the major storages. The Supplemental Appendix reports the estimated parameters (Table A1) and provides an illustration of how well the shape of the lognormal distribution approximates that of the estimated order-level valuations (Figure A1).

Figure 3: Illustration of uniform-price market clearing



6 Counterfactual analysis

6.1 Benchmark: batch uniform-price market clearing

Consider the theoretical benchmark of crossing latent supply and demand at discrete time intervals, as illustrated in Figure 3. A market clearing price, quantity, and allocation are determined by where the aggregate supply and demand curves cross in each period. Within each period, this arrangement would yield the efficient allocation. In particular, I will consider market clearing on a weekly basis for comparability with the observed exchange, where the mean duration of a limit order is about 8 days. In finance, call or batch auctions at discrete times have long been considered an alternative to continuous trading; Economides and Schwartz (1995), Handa and Schwartz (1996), and others have argued the merits of concentrating orders in a call auction. Also, Rustichini, Satterthwaite, and Williams (1994) and Satterthwaite and Williams (2002) show that trade at a single market-clearing price through a double auction makes the worst-case inefficiency converge to zero at the fastest possible rate as the number of traders increases.

This benchmark differs in important ways from the exchange observed in the data. First, by clearing at discrete time intervals (batching), the benchmark temporally aggregates traders. Relatedly, the benchmark generates a single market-clearing price in each period that sorts which buyers and sellers get to trade. Second, the benchmark performs aggregated exchange as defined in Section 2.4; per Figure 3, the benchmark

automatically splits supplied parcels to allocate units across multiple buyers who each demand idiosyncratic quantities. Finally, the theoretical benchmark crosses latent supply and demand, so there is no price shading by construction. To supplement the benchmark, which quantifies an ideal rather than a mechanism, I also consider a closely related incentive compatible mechanism adapting McAfee (1992).

I use the model primitives estimated per Section 5 to counterfactually simulate the benchmark discussed above. Aside from the change in exchange design to batch market clearing, I endeavor to keep all else the same as on the observed exchange. First, the same traders arrive at the same times, with the same quantities,²⁵ and with the same values $\hat{v}_i$ or costs $\hat{c}_i$ as on the observed exchange. Second, each trader still submits a single per-unit price for their order. Finally, traders are subject to the same percentage fee and same flat fee per traded parcel, but I now eliminate the flat fee duplication that would currently occur if a parcel were split across multiple counterparties. Indeed, because the benchmark market clearing process splits parcels automatically, traders would not even know and would not have a say in how many counterparties are involved in their trade. Implementation details follow; Section 6.2 presents the simulation results.

Implementation details In any given week, I observe buyers and sellers arrive and make a limit or market order on the observed exchange. For each observed limit order, I have an estimated $\hat{v}_i$ or $\hat{c}_i$ to use in simulating the counterfactual. For each observed market order, I know that $v_i \geq \hat{v}_i$ for buy orders and $c_i \leq \hat{c}_i$ for sell orders, where $\hat{v}_i$ and $\hat{c}_i$ are the relevant thresholds for choosing a market order. Therefore, for each buy market order, I randomly draw a v_i from the left-truncated distribution $[\hat{F}_b(\cdot|q_i, x_{w,i}) - \hat{F}_b(\hat{v}_i|q_i, x_{w,i})]/[1 - \hat{F}_b(\hat{v}_i|q_i, x_{w,i})]$, and for each sell market order, I randomly draw a c_i from the right-truncated distribution $\hat{F}_s(\cdot|q_i, x_{w,i})/\hat{F}_s(\hat{c}_i|q_i, x_{w,i})$. I now have a value or cost associated with every trader arrival. For ease of notation, I will refer to both estimated and simulated valuations collectively using the hat notation as $\hat{v}_i$ and $\hat{c}_i$. Given exchange and state fees, a buyer's market-facing value, or the per-unit price at which she breaks even, equals $(\hat{v}_i - f_b/q_i)/\tau_b$, while a seller's market-facing cost is $(\hat{c}_i + f_s/q_i)/\tau_s$.

Armed with market-facing valuations for all trader arrivals, I construct the ag-

²⁵Later, I also simulate the benchmark under the endogenous quantity model of Section 4.1, where counterfactual quantities can differ from the observed quantities.

gregate supply and demand curves for each calendar week. As each trader submits a single per-unit price for their parcel, the aggregate supply and demand curves are not smooth but involve steps. The market-clearing quantity (or total trade volume) is defined such that every unit for which the market-facing value of the buyer is greater than or equal to the market-facing cost of the seller is traded. Clearing the market in this way can result in partial execution of the quantity of up to one marginal trader per period. If this happens, I reduce the flat fee to $f \times \frac{\text{executed quantity}}{\text{order quantity}}$ only for the marginal trader as a special case, so that partial execution does not alter their market facing value or cost per unit. The Supplemental Appendix (A.4) elaborates further on implementing the benchmark.

6.2 Benchmark simulation results

It is informative to begin by examining the simulated benchmark at the trade level, comparing the composition of counterfactual trades to observed trades. A visual inspection of trade plots reveals that, on the observed exchange, relatively low-value demand units and high-cost supply units are able to trade by exploiting the time, price, and size dispersion available there. These trades are still individually Pareto-improving, but they may happen at the cost of more efficient potential trades involving higher-value demand or lower-cost supply.

Zooming out, Table 3 compares aggregate outcomes in the benchmark versus the observed exchange. To make this comparison under various models of trader behavior, the table displays each respective $\frac{\text{counterfactual outcome} - \text{observed outcome}}{\text{observed outcome}} \times 100\%$. The primary outcome of interest is total trade surplus, defined as the sum of buyers' profit $\hat{v}_i - f_b/q_i - \tau_b p_i$ and sellers' profit $\tau_s p_i - f_s/q_i - \hat{c}_i$ for all traded units of water.

The benchmark achieves higher trade surplus than the observed exchange; the size of the gap varies with the model of trader behavior. Consider first the outcomes given $\eta = 0.99$ in row (a). Recall that there would be very little price shading by traders on the observed exchange given this level of risk aversion, so price shading is effectively excluded as a possible source of discrepancy with the benchmark. Table 3 reports that trade surplus would still be 28% higher in the benchmark. Interestingly, total trade volume does not change much in row (a), so this surplus improvement does not rely on expanded trade. The benchmark simply allocates water units more efficiently than the observed exchange such that the surplus per unit traded is 27%

Table 3: Percent change of counterfactual outcomes from observed exchange, Zone 7

	CRRA η	counterfactual	total trade surplus	surplus per unit traded	total trade volume
<i>Baseline model</i>					
(a)	0.99	benchmark	+28%	+27%	+1%
(b)	0.57	benchmark	+44%	-22%	+85%
(c)	0.57	dominant strategy double auction	+40%	-19%	+73%
<i>Endogenous quantity model</i>					
(d)	0.99	benchmark	+38%	+32%	+5%
(e)	0.57	benchmark	+52%	-19%	+87%

Notes: Table reports the percent change in simulated counterfactual outcomes from corresponding outcomes of the observed exchange. The product is water sourced from Zone 7. The time period is the full sample period excluding times when the Goulburn to Murray trade limit was open. Section 6.5 analyzes this trade limit and interzone trade.

higher.

Next, consider the $\eta = 0.57$ model in row (b). The observed exchange features non-negligible price shading at this risk aversion level, as quantified in Section 5.3. Price shading can block potential trades where the trade surplus is small relative to the size of markups or markdowns. Table 3 indicates there are many such marginal trades; the benchmark is able to expand total trade volume considerably by picking them up. Because the trade surplus of these marginal trades is relatively small, the average surplus per unit traded is smaller in the benchmark than on the observed exchange. On balance, the surplus gains of the benchmark, at 44%, are larger when the observed exchange exhibits price shading than when it does not as in row (a).

To supplement the benchmark, which is an ideal rather than a mechanism, I also simulate a closely-related incentive compatible mechanism by adapting the dominant strategy double auction proposed by McAfee (1992).²⁶ Incentive compatibility, i.e., making it incentive compatible for traders to truthfully report their values, comes at the cost of a surplus loss relative to the benchmark. Table 3, row (c) displays the results of the double auction for $\eta = 0.57$. (For $\eta = 0.99$, traders already post $p \approx v_{mkt}$ due to their risk aversion; a dominant strategy auction is not separately needed to induce truthtelling.) Compared with corresponding results of the benchmark in row (b), the percent changes of the dominant strategy auction are of reduced but similar magnitude; the total trade surplus gain over the status quo would be 44%

²⁶The original McAfee (1992) auction is for traders demanding or supplying one unit each. I make adjustments to maintain incentive compatibility in a multi-unit setting.

in the benchmark versus 40% in the dominant strategy auction. Thus, incentive compatibility does not constitute the fundamental gap between the benchmark and the observed exchange. See Supplemental Appendix Section A.3 for more about the dominant strategy double auction and its implementation.

Per Section 6.1, the analysis here limits counterfactual trader arrivals to actual trader arrivals as observed during the timeframe of analysis. In this sense, the numbers reported in Table 3 are conservative in that they may understate the gains from better exchange design. Higher expected trade surplus could draw more participants, increasing liquidity and further increasing exchange surplus. We can indeed simulate the additional impact of increased arrivals; this would, however, rely on tentative speculation about the population of relevant potential traders, especially as the exchange data are anonymous and unlinked.

6.3 Robustness to endogenous quantities

Next, I simulate the counterfactual benchmark under the endogenous quantity model of Section 4.1, where each trader has an individual supply or demand curve given by equation (12). Table 3, rows (d)-(e) provide the endogenous quantity outcomes corresponding to rows (a)-(b) from the baseline model. The endogenous quantity model yields stronger results; that is, the surplus gap between the benchmark and the observed exchange is even larger than in the baseline model. The richer expression of traders' supply and demand curves, or valuations at alternative quantities, alleviates potential quantity distortions and allows the benchmark to further improve exchange surplus. Meanwhile, the baseline model yields more conservative predictions regarding counterfactual improvements, which is a desirable feature from a policy perspective. Details of simulating the benchmark under the endogenous quantity model follow.

Implementation details Under the endogenous quantity model, the benchmark now has traders expressing marginal market-facing values or costs for each unit they demand or supply. Thus, the counterfactual trade quantity of a given trader can differ from their observed order quantity, depending on where the market-clearing price falls. To avoid artificially inflating supply beyond what is actually available, I do not allow sellers to supply more than they did on the observed exchange. Similarly aiming for conservatism, I allow buyers to express valuations for up to ten percent more than they ordered on the observed exchange.

Traders remain subject to the same percentage fee and flat fee implemented in the baseline benchmark of Section 6.1. Note that, by definition, a flat fee is charged as soon as even one unit trades. So each trader wants to trade a sufficient number of units at once, or none at all, to absorb the flat fee across multiple units. This generally leads to one uniformly-priced step at the bottom of each supply curve or top of each demand curve, even when traders can freely express discriminatory valuations for each unit. For such a step with t units, a seller expresses a cost of (average cost of the t units $+ f_s/t$)/ τ_s per unit; for each remaining unit outside of the step, she expresses the respective marginal cost/ τ_s . The size of the step equals $\min\{t \mid \text{average cost of } t \text{ units} + f_s/t < \text{marginal cost of unit } t+1\}$. Likewise, buyers size the step to equal $\min\{t \mid \text{average value of } t \text{ units} - f_b/t > \text{marginal value of unit } t+1\}$. This is an interesting insight regarding uniformly-priced steps in multi-unit exchange: fixed transaction costs can contribute to their formation. If the marginal traded unit is in one of these steps, the flat fee is adjusted to $f \times \frac{\text{executed quantity}}{\text{step quantity}}$ only for the marginal trader as a special case.²⁷

As shown in Section 4.1, the curvature and scale parameters of each trader's demand or supply curve are point-identified for continuous quantity choice, while we obtain bounds in the case of discrete quantity choice. Discrete cases include market orders as well as limit orders for which the probability of order execution is not locally differentiable in quantity. This occurs in some time-period-by-zone combinations where quantities in multiples of 10ML exhibit statistically significant jumps in execution probability. For counterfactual simulation purposes, it is necessary to assign specific demand/supply curve parameters, from within the identified bounds, to each such case that lacks point identification. I assign the curvature parameter b_i predicted by random forest regression given the relationship between point-identified $\hat{b}_i$ and $\{q_i, v_i(q) \text{ or } c_i(q), \text{ the year, and the } \textit{waterday} \text{ of order placement}\}$. If the predicted parameter falls outside the bounds, I assign $\max(\min(\text{prediction}, \text{upper bound}), \text{lower bound})$. Given b_i and $v_i(q) \text{ or } c_i(q)$, the corresponding scale parameter a_i is determined.

Lastly, as discussed at the end of Section 4.1, there are reasons to expect some order quantities to be exogenous with respect to the exchange. Generally, these

²⁷This is analogous to the fee applied to the marginal trader in the baseline benchmark (see Section 6.1), with the same rationale, but replacing order quantity with step quantity in the denominator to avoid an incentive to submit extra units at unserious prices. Order quantity and step quantity are the same in the baseline benchmark because traders submit a single price for all units.

observations would not be rationalizable by a model of exchange-optimized q with smoothly elastic supply or demand curves. Therefore, I take a data-driven approach to flag observations unlikely to be generated by my model of endogenous q , and I treat quantities as exogenous in these instances. In particular, I flag observations that would imply decreasing seller marginal costs, increasing buyer marginal values, a negative curvature parameter (which would imply negative marginal valuations), or where the lower bound in (14) would exceed the upper bound in (13).

6.4 Interaction with drought

Table 4 subdivides the data sample into shorter periods and reports period-specific outcomes of the benchmark. Each period roughly corresponds to a quarter of a year. The table excludes dates during which the Goulburn to Murray trade limit was open, such as period 4 that falls between periods 3 and 5; I devote Section 6.5 to explaining the trade limit and analyzing period 4 specifically. In Section 2.3, Figure 1, the solid line shows that the driest period in the sample was spring 2020, corresponding to time period 1, while period 5 was the wettest. Improved exchange performance would be particularly welcome and important during drought.

Comparing period 1 outcomes in Table 4 to the overall outcomes in Table 3, we see that the benchmark would generate especially large improvements to trade surplus during the driest period, when such improvements are most critical. One driver behind this is that the benchmark’s impact on surplus per unit traded tends to be higher in the driest period; see Table 4, column (b). This comes from value heterogeneity being especially high during droughts, so that there are higher per-unit gains to be had from sorting the highest-value demand units and lowest-cost supply units into trade, as the benchmark does relative to the observed exchange. Rafey (2023) makes a related but distinct observation that gains from the existence of water trade, versus no trade, are largest during droughts. Here, I add the insight that, even given trade exists, and even with status quo gains from trade already being larger during drought, better exchange design further improves those gains by an especially high percentage during drought.

To elaborate on this point, Figure 4 visualizes the fact that value heterogeneity is especially high during droughts. It plots buyers’ market-facing values and sellers’ market-facing costs at the order level, including orders that did not execute, at two

Table 4: Percent change of benchmark outcomes from observed exchange, Zone 7

time period	(a) total trade surplus	(b) surplus per unit traded	(c) total trade volume
<i>Baseline model, CRRA $\eta = 0.99$</i>			
1. Apr 1, 2020–May 30, 2020	+126%	+116%	+5%
2. Jul 1, 2020–Oct 3, 2020	+5%	+20%	-13%
3. Oct 4, 2020–Jan 30, 2021	+10%	+18%	-7%
5. Jul 1, 2021–Oct 2, 2021	+29%	+3%	+25%
<i>Baseline model, CRRA $\eta = 0.57$</i>			
1. Apr 1, 2020–May 30, 2020	+62%	+26%	+29%
2. Jul 1, 2020–Oct 3, 2020	+19%	-18%	+46%
3. Oct 4, 2020–Jan 30, 2021	+30%	-14%	+52%
5. Jul 1, 2021–Oct 2, 2021	+128%	-33%	+238%

Notes: Table reports the percent change in simulated counterfactual outcomes from corresponding outcomes of the observed exchange. The product is water sourced from Zone 7.

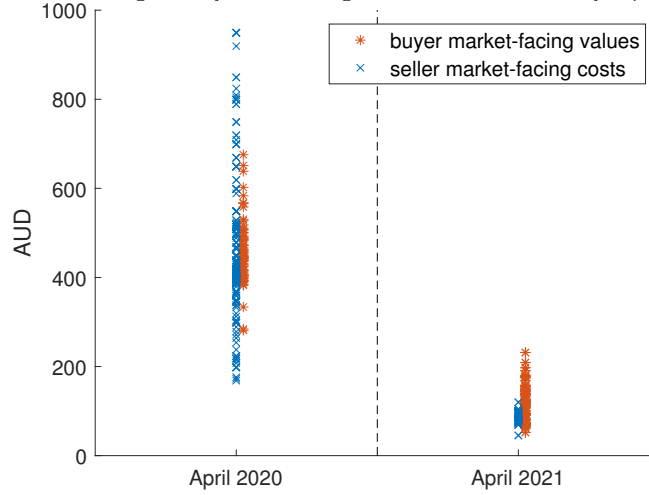
different points in time. The first is April 2020, when water availability was lowest in my sample, and the second is the same month a year later in April 2021, when water was plentiful in comparison. Indeed, the support of traders' values is much wider in April 2020. As a result, rearranging these units into more efficient trades generates sizable surplus gains. For example, the benchmark would have reduced the average opportunity cost of traded water $\hat{c}$ to about 203 AUD/ML in April 2020, compared to 239 AUD/ML on the observed exchange.

Meanwhile, wet periods like period 5 feature low value heterogeneity and generally lower per-unit surplus gains to be had from more efficient trade; see Table 4, column (b). Relatedly but distinctly, a large proportion of potential trades may be marginal when water is abundant, meaning the value difference between the potential buyer and seller is small. In that scenario, the differential ability of a more efficient exchange to pick up marginal trades can contribute to large increases in relative trade volume, as in Table 4, period 5-column (c).

6.5 Exchange design versus liquidity expansion

The analysis in Sections 6.2-6.4 reveals that exchange design can make a difference, but one may wonder how important matters of design are relative to other changes

Figure 4: Value heterogeneity in drought versus in plenty, $\eta = 0.99$, Zone 7



or interventions in the market, such as increased liquidity. In this section, I leverage the opening of the Goulburn-to-Murray trade limit, described below, to shed some light on this question.

Australian water authorities use trade limits to regulate the physical impacts of intervalley trade. Trade limits are either open or closed at a moment in time, allowing or disallowing trade from one area to another. In particular, the Goulburn-to-Murray trade limit did not allow trade from the Goulburn and some other upstream systems down to the River Murray if the total water owed to the Murray exceeded a cap at any one time. During my data sample and the years immediately before, the trade limit was closed more often than not. As a consequence, water could not be sold from sellers in Zone 1A to buyers in Zone 7 for most of the sample period. While trade in the opposite direction, from Zone 7 to 1A, was permitted during trade limit closures, it was negligible in practice due to price differentials between the zones. Thus, the two most traded allocation products were effectively isolated from each other much of the time.

In my data sample, the main openings of the Goulburn-to-Murray trade limit were in June 2020, intermittently in February and March 2021, and an extended continuous opening from March 18 to June 30, 2021,²⁸ which were excluded from the Section 6.2-6.4 analyses. When the trade limit is open, Zone 7 and 1A can trade with each other freely; that is, there are no additional trade costs or barriers between these

²⁸Other brief openings included May 24, 2020 (for 14 minutes); July 1, 2020; and July 1, 2021 (for 12 minutes).

zones, which are located in the same state.²⁹ Thus, during March 18–June 30, 2021, the two zones were like one large market, with expanded liquidity compared to the counterfactual of isolated markets during the same time.

I use the period March 18–June 30, 2021, or “period 4”, to assess the relative impact of exchange design as follows. First, I simulate a counterfactual in which Zone 7 and 1A are isolated from each other, allowing no trade between them, but in which the respective products are traded on the benchmark exchange described in Section 6.1. This is in contrast to the actual trades observed during period 4, which enjoyed open interzone trade but were arranged by the status-quo exchange. Next, I compare the counterfactual period-4 trades to the actual trades, in terms of combined trade surplus from the two products, water sourced from Zone 7 and 1A.

Despite the disadvantage of isolated markets, I find that the counterfactual trades yield 1.02 times the trade surplus of the observed trades under the baseline model at $\eta = 0.99$; with $\eta = 0.57$, they yield 1.13 times the trade surplus of the observed trades. Thus, the impact of exchange design simulated here is comparable to, or possibly exceeds, that of a liquidity expansion event that opened trade between the two biggest zones by trade count. I interpret this to mean that the design of how trades are arranged is important, not only in an absolute sense but also relative to other market interventions. Moreover, even where environmental and hydrological realities constrain some types of reforms, exchange design can achieve efficiency improvements by making the most of trade given the boundaries imposed.

6.6 A decomposition for design reform

So far, we have seen that there is room to improve this market through exchange design, exchange design matters both in an absolute and relative sense, and the impact would be especially large during drought. At the same time, the benchmark of batch market clearing would constitute a drastic change in operation from the status quo. For the sake of practical policy, could there be found an incremental reform that still makes a sizable difference? To answer this question, it is useful to decompose the gap between the observed exchange and the counterfactual benchmark, separating the contributions of parcel splitting, price shading, and batching. I conduct additional

²⁹Allocation trade between them is processed through a series of water accounting adjustments at the individual account and bulk level; individual trades do not cause immediate physical movement, as river managers make bulk releases to meet aggregate forecasted demand. See Section 2.1.

Table 5: Percent change from observed exchange, total trade surplus, Zone 7

	$\eta = 0.99$	$\eta = 0.57$
(a) observed exchange	0%	0%
<i>Counterfactual simulations, baseline model:</i>		
(b) + parcel splitting (price as observed)	+11%	+22%
(c) + parcel splitting + no price shading	–	+25%
(d) + parcel splitting + no price shading + batching = benchmark	+28%	+44%

Notes: Table reports the percent change in total trade surplus of each simulated counterfactual from that of the observed exchange. The product is water sourced from Zone 7. The time period is the full sample period excluding times when the Goulburn-to-Murray trade limit was open.

simulations to isolate the effect of each feature.

First, I simulate a new counterfactual that maintains the same limit order prices and continuous-time nature as the observed exchange, but splits parcels. Specifically, I implement 1 ML unit splits of the sellers’ parcels so that each buy order executes against the aggregate supply curve. This lets each buyer fill their order quantity with the cheapest units currently available on the exchange, regardless of how many different sellers these units come from and without having to buy the entirety of any seller’s parcel. To respect potential lumpiness in buyers’ demand and to understate rather than overstate the counterfactual gains, I let buyers buy only if they can fill their entire order quantity with units that are cheaper than their value; if the supply of such units is less than a buyer’s order quantity, she does not buy at all. Noting that over 90% of observed trades are generated by buyers fulfilling seller limit orders, I simplify the simulation by having sellers submit limit orders only and buyers submit market orders only. As for fees, they are implemented as in the benchmark, so that parcel splitting does not result in duplication of the flat fee; see Section 6.1. If a seller’s limit order executes partially, their flat fee is reduced to $f \times \frac{\text{executed quantity}}{\text{order quantity}}$.

Next, I simulate a counterfactual exchange that additionally eliminates price shading. Here, sellers’ limit order prices are set equal to their market-facing costs, while other details of the simulation are the same as in the previous paragraph. The difference between the previous paragraph’s simulation and this one isolates the contribution of price shading. Finally, the gap from this simulation to the benchmark isolates the contribution of batching.

Table 5 reports how the simulated total trade surplus generated by these counter-

factual scenarios differs from that of the observed exchange. The gap from row (a) to row (b) isolates the contribution of freely splitting parcels. The parcel splitting in (b) closes about 40-50% of the total surplus gap from the observed exchange to the benchmark, (d)–(a). From a practical point of view, it is useful to learn that the simple act of parcel splitting would go a long way towards making trade more efficient, all while preserving a continuous-time market. Next, the difference between rows (b) and (c) isolates the contribution of price shading. Note that row (c) is left blank for $\eta = 0.99$ because there is effectively no price shading with that level of risk aversion. Lastly, the gap from row (c) to row (d) isolates the contribution of batching, which consolidates each week’s supply and demand and sorts the highest-value buyers and lowest-cost sellers into trade. Table 5 suggests this element is relatively more important if there is less price shading, such as if $\eta = 0.99$; for a given set of observed limit order prices, less price shading implies higher seller costs and lower buyer values, and efficient buyer-seller sorting matters more in such an environment.

A robust practical takeaway from this analysis is that free parcel splitting would be an effective incremental reform to target. Reforming the fee structure, so that trade fees no longer depend on the number of counterparties amongst which a parcel is split or from which a buy order is sourced, is prerequisite. The recommended design change is conceptually simple, preserves the existing continuous market environment, and makes a big dent in the gap to the benchmark ideal.

7 Conclusion

This paper studies how the trading mechanism affects water exchange outcomes using exchange data from Australia, a world leader in market-based water management. In this novel application of multi-unit exchange, I assess the impact of exchange design through a structural analysis of the buy and sell orders and trades. Specifically, I combine the data with models of participants’ best-response actions on the exchange to identify the latent values underlying the orders. I conduct this analysis under different models and with data from different trading zones to gain a robust understanding from the collective body of results. Methodologically, I make new contributions to the analysis of limit order markets by endogenizing order quantity in addition to order price and incorporating behavioral flexibility suitable for water market participants.

The analysis in this paper finds that there is ample room to improve the market

through exchange design, exchange design matters both in an absolute and relative sense, and the impact would be especially pronounced during drought, when value heterogeneity is highest and the costs of misallocation are most severe. Investigating the features of the status quo exchange that could be responsible for the performance gap, I discover that parcel (in)divisibility is a prime target for practical reform, as it accounts for close to half of the gap.

There is much more work to be done on water markets by economists. For example, data that links exchange order details to the population of farms would enable comprehensive, front to back modeling of their water management behavior. While the Australian government does not have that linkage yet, future improvements to data systems could allow farm-to-exchange analysis. Also, this paper studies water exchange in a region where the history, hydrology, and institutions were conducive to an early emergence of formal water markets. An important, ongoing question is how to design markets in regions like the American West where water rights laws and hydrological conditions differ and have so far been less conducive to flourishing trade.

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A Supplemental Appendix to “Liquid Markets: An Empirical Analysis of a Water Exchange”

Yunmi Kong

A.1 Proofs

Assumption 1S. *The execution probability $P_s(\cdot, \cdot)$ for a sell limit order satisfies*

- (i) *Monotonicity: $P_s(p, q)$ is strictly decreasing in p .*
- (ii) *Decreasing reversed hazard rate: $P_s(p, q)/P_{s,1}(p, q)$ is strictly increasing in p .*
- (iii) *Uncertain execution: $P_s(p, q) < 1$.*

Proof of Proposition 1

Proof. The proof proceeds by showing that $0 < \frac{\partial L(v)}{\partial v} < \frac{\partial M(v, \ell)}{\partial v}$ for all $v > \bar{\ell}$, where $\bar{\ell} \equiv \tau\ell + f/q$. If this is true, there will be at most one value $\tilde{v}(\ell) > \bar{\ell}$ at which $L(v) = M(v, \ell)$, with $L(v) > M(v, \ell)$ for $v \in (\bar{\ell}, \tilde{v}(\ell))$ and $L(v) < M(v, \ell)$ for $v \in (\tilde{v}(\ell), \infty)$. By the envelope theorem, $\frac{\partial L(v)}{\partial v} = P(p^*(v, q), q)$. We have $0 \leq P(p^*(v, q), q) < 1$ by definition of a probability and Assumption 1. Thus, $0 \leq \frac{\partial L(v)}{\partial v} < 1 = \frac{\partial M(v, \ell)}{\partial v}$. This completes the proof.

Note that the threshold $\tilde{v}(\ell)$ will be in $(\bar{\ell}, p^{*-1}(\ell; q))$. This is because, first, buyers with $v \leq \bar{\ell}$ definitely prefer a limit order because $M(v, \ell) \leq 0 \leq L(v)$ for $v \leq \bar{\ell}$. Second, for v such that $p^{*-1}(\ell; q) \leq v \Leftrightarrow \ell \leq p^*(v, q)$, buyers definitely prefer a market order with price ℓ because it guarantees execution at a weakly cheaper price than the optimal limit order. $\square$

Proof of Proposition 2

Proof. The only part of the identification argument not fully explained in the main text is the nonparametric identification of $F_b(\cdot|\cdot)$ and $F_s(\cdot|\cdot)$ given $\alpha_b > 0$ and $\alpha_s > 0$. The proof of that step follows. For simpler notation, fix q and consider $F(\cdot)$ as shorthand for $F(\cdot|q)$.

First, consider the buyers. Let the value threshold $\tilde{v}$, determining whether a rational buyer chooses a limit order or a market order, be fixed at an arbitrary value

t . Buyers with values $v \geq \tilde{v} = t$ will either rationally submit a market order or submit a limit order by default. Thus,

$$1 - F_b(t) = \frac{\#\{\text{buy market orders}|\tilde{v} = t\} + \#\{\text{buy limit orders with } v \geq t|\tilde{v} = t\}}{\#\{\text{all (market and limit) buy orders}|\tilde{v} = t\}}.$$

Having identified $F_b(t)$ using this ratio of observables, $F_b(\cdot|v \geq t) \equiv \frac{F_b(\cdot) - F_b(t)}{1 - F_b(t)}$ is nonparametrically identified as the empirical cdf of all the values associated with $\{\text{buy limit orders with } v \geq t|\tilde{v} = t\}$ —this is the set of limit orders submitted by buyers that default to limit orders, allowing us to identify the shape of $F_b(\cdot)$ for values above t . Also, $F_b(\cdot|v < t) \equiv \frac{F_b(\cdot)}{F_b(t)}$ on $[0, t)$ is nonparametrically identified as the empirical cdf of all the values associated with $\{\text{buy limit orders with } v < t|\tilde{v} = t\}$. Since $F_b(t)$ is identified above, identification of $F_b(\cdot|v \geq t)$ and $F_b(\cdot|v < t)$ implies identification of $F_b(\cdot)$ over its support.

Next, consider the analogous argument for sellers. Let the cost threshold $\tilde{c}$, determining whether a rational seller chooses a limit order or a market order, be fixed at an arbitrary t . Sellers with costs $c \leq \tilde{c} = t$ will either rationally submit a market order or submit a limit order by default. Thus,

$$F_s(t) = \frac{\#\{\text{sell market orders}|\tilde{c} = t\} + \#\{\text{sell limit orders with } c \leq t|\tilde{c} = t\}}{\#\{\text{all (market and limit) sell orders}|\tilde{c} = t\}}.$$

Having identified $F_s(t)$ using this ratio of observables, $F_s(\cdot|c > t) \equiv \frac{F_s(\cdot) - F_s(t)}{1 - F_s(t)}$ is nonparametrically identified as the empirical cdf of all the values associated with $\{\text{sell limit orders with } c > t|\tilde{c} = t\}$. Also, $F_s(\cdot|c \leq t) \equiv \frac{F_s(\cdot)}{F_s(t)}$ on $[0, t)$ is nonparametrically identified as the empirical cdf of all the values associated with $\{\text{sell limit orders with } c \leq t|\tilde{c} = t\}$ —this is the set of limit orders submitted by sellers that default to limit orders, allowing us to identify the shape of $F_s(\cdot)$ for costs below t . Since $F_s(t)$ is identified above, identification of $F_s(\cdot|c > t)$ and $F_s(\cdot|c \leq t)$ implies identification of $F_s(\cdot)$ over its support. $\square$

A.2 Limit order pricing with revision

In this section, I allow for the possibility that a trader will revise her limit order after placing it. The purpose is to adjust for the (possible) continuation value of revision when identifying traders' latent values. I abstract from conditioning the model on

covariate vector x and introduce the model from the buyer's perspective.

After the passage of an exogenous, stochastic interval of time since initial order placement, the buyer will revisit her limit order and, if it has not yet executed, exit the market with exogenous probability $1 - r$ or else revise the price with probability r . Empirically, I flag potential revisions as instances in which a limit order ends unexecuted and subsequently a new limit order appears on the same day with the same quantity and a different price. Motivated by this data, in which less than 1.5 percent of unexecuted limit orders are revised more than once, I model traders revising their limit order up to one time. Then the buyer's problem of choosing a price in the last revision stage is identical to (1). The model is generalizable to any fixed, finite maximum number of revisions by iterating the revision model up to that number and solving backwards. The buyer's per-unit value for water at the time of revision is denoted $v_{r,i}$. This $v_{r,i}$ has conditional distribution $F^r(\cdot|v_i, q_i)$ which is weakly stochastically ordered in initial value v_i so that a buyer with higher v_i is weakly more likely to have a higher $v_{r,i}$. Persistent values, $v_{r,i} = v_i$, would be a special case of this more general setup.

Assumption 3. *Distribution $F^r(\cdot|v_i, q_i)$ of $v_{r,i}$ is weakly stochastically ordered in v_i .*

Let $C(v_i, q_i, p_i)$ denote the expected profit per unit from revising a limit order in the future, as perceived by the buyer at the time she submits the initial limit order. The arguments v, p are those pertaining to the *initial* limit order. This $C(v_i, q_i, p_i)$ is analogous to a continuation value and will be an input to the buyer's optimization problem of choosing limit order price p . Mathematically, define $C(v_i, q_i, p_i)$ as

$$C(v_i, q_i, p_i) \equiv E_{v_{r,i}} \left[P(p_r^*(v_{r,i}, q_i), q_i) (v_{r,i} - \tau p_r^*(v_{r,i}, q_i) - f/q_i) \middle| v_i, q_i, p_i \right]. \quad (\text{A.1})$$

The buyer's optimal revision price as a function of her value $v_{r,i}$ is denoted $p_r^*(v_{r,i}, q_i)$, and $P(p_r^*(v_{r,i}, q_i), q_i)$ is the probability that a buy limit order with price $p_r^*(v_{r,i}, q_i)$ executes.

Assumption 4. *Regarding the expected profit $C(v_i, q_i, p_i)$ of revising a limit order,*

- (i) *The buyer approximates that $C(v, q, p) = C(v, q)$.*
- (ii) *$r|C(v', q) - C(v, q)| < |v' - v|$ for all $v' \neq v$.*

Assumption 4-(i) means the buyer approximates that her own choice of limit order price p will not influence the evolution of the market to such an extent that it would

affect her revision payoff in the future. This is similar to a common assumption made when studying dynamic auctions on eBay, in which a buyer's continuation value is assumed not to depend on her previous actions; see, e.g., Backus and Lewis (2025), Bodoh-Creed et al. (2021), and Hendricks et al. (2025). One rationale in the case of eBay is that the influence of any one bid by one buyer dissipates quickly over time. Another rationale is that it represents realistic buyer behavior because forecasting the evolution of the entire market in response to one's individual action is complex while the gains of doing so for the buyer are small.

Assumption 4-(ii) means the expected profit of future revision, multiplied by the probability of revision r , is less sensitive to the argument v than is v itself. It is satisfied in both of the following models of the relationship between v_i and $v_{r,i}$. First, it is automatically satisfied if $v_{r,i}$ and v_i are conditionally independent with $F^r(v_r|v, q) = F^r(v_r|q)$. In this scenario, $C(v, q)$ does not depend on v so $C(v', q) - C(v, q) = 0$. Second, the assumption is also satisfied in the opposite scenario of perfectly persistent values, $v_{r,i} = v_i$.¹ In general, satisfaction of Assumption 4-(ii) depends on the function $P(\cdot, \cdot)$ and the joint distribution of (v, v_r) . The next corollary follows from the previous assumptions.

Corollary 1. $\partial C(v, q)/\partial v \geq 0$.

Now, consider a buyer's choice of price p for a limit order. Given her value v_i , buyer i chooses price p_i to maximize her expected utility from the order in light of the execution probability $P(\cdot, \cdot)$ and accounting for the possibility of future revision. Her maximization problem is

$$\max_p P(p, q_i)[(v_i - \tau p)q_i - f] + (1 - P(p, q_i))rC(v_i, q_i)q_i. \quad (\text{A.2})$$

If the buyer's limit order is executed, she gets payoff $(v_i - \tau p)q_i - f$, and if it is not executed, she revises the order with probability r , yielding expected payoff $C(v_i, q_i)q_i$ from the revision. Differentiating this objective function with respect to p , the first-order condition yields

$$v_i - rC(v_i, q_i) = \left(\tau p_i + \frac{f}{q_i} \right) + \tau \frac{P(p_i, q_i)}{P_1(p_i, q_i)}. \quad (\text{A.3})$$

¹In this scenario, the expression for $C(v, q)$ simplifies to $P(p_r^*(v, q), q)(v - \tau p_r^*(v, q) - f/q)$, so by the envelope theorem, $\partial C(v, q)/\partial v = P(p_r^*(v, q), q) < 1$, where the inequality follows from Assumption 1-(iii). Then $r|\int_{y=v}^{v'} \frac{\partial C(y, q)}{\partial y} dy| < |\int_{y=v}^{v'} 1 dy|$, satisfying Assumption 4-(ii).

This first-order condition is similar to equation (6), but the buyer now chooses a price in light of a pseudo value that equals her value v_i minus the continuation value $rC(v_i, q_i)$. In other words, since the buyer's expected payoff from an unexecuted order is now positive, she will behave as if her value is lower by that amount when choosing a limit order price. Under Assumptions 1 and 4, a buyer's optimal choice of limit order price p is strictly increasing in her value v .

Corollary 2. *The buyers' best-response price function $p^*(v, q)$ for a limit order is strictly increasing in v .*

Seller analogs Now consider the equivalent problem for a seller, in which the seller analogs of Assumptions 3 and 4 apply. Let the seller's per-unit opportunity cost of relinquishing water at the time of revision be denoted $c_{r,i}$. Meanwhile, let $C_s(c_i, q_i)$ denote the expected profit from revising a limit order in the future, as perceived by the seller at the time she submits the initial limit order: $C_s(c_i, q_i) \equiv E_{c_{r,i}} \left[P_s(p_r^*(c_{r,i}, q_i), q_i) (\tau_s p_r^*(c_{r,i}, q_i) - c_{r,i} - f_s/q_i) \mid c_i, q_i \right]$. When choosing a price p for such a limit order, her maximization problem is $\max_p P_s(p, q_i) [(\tau_s p - c_i) q_i - f_s] + (1 - P_s(p, q_i)) r C_s(c_i, q_i) q_i$. Differentiating this objective function with respect to p , the first-order condition yields

$$c_i + r C_s(c_i, q_i) = \left(\tau_s p_i - \frac{f_s}{q_i} \right) + \tau_s \frac{P_s(p_i, q_i)}{P_{s,1}(p_i, q_i)}. \quad (\text{A.4})$$

Compared to (7), the seller behaves as if her cost is higher by the amount $r C_s(c_i, q_i)$. Corrollaries 1S and 2S are seller analogs of Corrollaries 1 and 2.

Corrollary 1S. $\partial C_s(c, q) / \partial c \leq 0$.

Corrollary 2S. *The sellers' best-response price function $p^*(c, q)$ for a limit order is strictly increasing in c .*

Identification First, the buyer's value v_r or seller's cost c_r associated with each observation of a *revised* limit order is identified by (6) and (7). Next, using the monotonicity in Corrollaries 2 and 2S, I define new notation $\tilde{C}(\cdot, \cdot)$ for the expected profit of revising a limit order such that $\tilde{C}_b(p^*(v, q), q) = C_b(v, q)$ for a buyer and $\tilde{C}_s(p^*(c, q), q) = C_s(c, q)$ for a seller. This change of variable replaces the unobserved latent value with observable p . In detail, it means replacing $F_b^r(v_r | v, q)$ with

$\tilde{F}_b^r(v_r|p^*(v, q), q)$ inside the definition of $C_b(\cdot, \cdot)$. For a buyer, this gives $\tilde{C}_b(p, q) \equiv \int_{v_r} P_b(p_r^*(v_r, q), q)(v_r - \tau_b p_r^*(v_r, q) - f_b/q) d\tilde{F}_b^r(v_r|p, q)$. As $P_b(\cdot, \cdot)$ and the value v_r associated with each observed revision price $p_r^*(v_r, q)$ are identified, $\tilde{C}_b(p, q)$ is identified. Analogous logic applies to identification of the seller's $\tilde{C}_s(p, q)$.

Having identified this $\tilde{C}_b(\cdot, \cdot)$ and $\tilde{C}_s(\cdot, \cdot)$, the buyer's v or seller's c associated with an *initial* limit order is identified by replacing $C_b(v, q)$ and $C_s(c, q)$ with $\tilde{C}_b(p, q)$ and $\tilde{C}_s(p, q)$ in (A.3) and (A.4):

$$v_i = \left(\tau_b p_i + \frac{f_b}{q_i} \right) + \tau_b \frac{P_b(p_i, q_i)}{P_{b,1}(p_i, q_i)} + r \tilde{C}_b(p_i, q_i), \quad (\text{A.5})$$

$$c_i = \left(\tau_s p_i - \frac{f_s}{q_i} \right) + \tau_s \frac{P_s(p_i, q_i)}{P_{s,1}(p_i, q_i)} - r \tilde{C}_s(p_i, q_i). \quad (\text{A.6})$$

A.3 A dominant strategy double auction

To supplement the benchmark, I also simulate a closely-related incentive compatible mechanism by adapting the dominant strategy double auction proposed by McAfee (1992). In this double auction, prices are uniform for each side and determined by the bids of the buyer and seller who marginally disqualify for trade. Thus, a trader's bid does not affect her price conditional on trade but only affects whether she trades. By the usual Vickrey argument (as applied in, e.g., a second-price auction), bidding one's true valuation is a dominant strategy in this auction. However, relying on excluded bids to determine prices leads to, in some situations, marginal trades being dropped, so incentive compatibility comes at the cost of a surplus loss relative to the benchmark. Also, the mechanism sometimes needs to pay sellers a lower price than it collects from buyers and makes money as a result.

The McAfee (1992) auction is for traders demanding or supplying one unit each. In my multi-unit setting, there is an additional incentive to bid untruthfully arising from the possibility of partial execution. If the market-clearing quantity results in partial execution of the marginal qualifying bid, then there are potential gains to be had from lying to move up in rank among trade-qualified bids. To restore incentive compatibility, I remove the link between bid rank and portion executed by executing an equal fraction of all trade-qualifying bids on the same side of trade. For interested readers, the next paragraph specifies further details of the multi-unit dominant strategy double auction that I implement. Aside from the mechanism, all

other implementation details remain the same as in Section 6.1.

In each calendar week, buyers report their per-unit value b , and sellers report their per-unit cost s . The mechanism will charge buyers a price p_b and pay sellers p_s per traded unit, to be determined below. Let demand units be ranked, highest per-unit value first, and let supply units also be ranked, lowest per-unit cost first. In auction version (A), let buyer k be the worst-ranked buyer for whom all demanded units have a value greater than or equal to the costs of correspondingly ranked supply units. Let seller $j(k)$ be the counterpart seller for k 's last unit. If next-ranked buyer $k + 1$ and seller $j(k) + 1$ exist, define $p_0 = \frac{1}{2}(b_{k+1} + s_{j(k)+1})$. Then,

$$\begin{cases} \text{if } b_{k+1} \leq s_{j(k)+1} \text{ and } p_0 \in [s_{j(k)}, b_k], & p_b = p_s = p_0; \\ \text{otherwise, if } b_{k+1} \geq s_{j(k)+1}, & p_b = b_{k+1}, p_s = s_{j(k)+1}. \end{cases}$$

In either of the above scenarios, buyers 1 through k will buy their demanded units in full, and sellers 1 through $j(k)$ will sell, with any partial execution applied as an equal fraction across them. If neither of the above scenarios applies, I move up the ordered buyer list in sequence until a buyer $k - i$ is found such that $b_{k-i+1} \geq s_{j(k-i)+1}$. Then $p_b = b_{k-i+1}$, $p_s = s_{j(k-i)+1}$, and buyers 1 through $k - i$ will buy their demanded units in full. If this condition is not met by any i , no trade occurs. In auction version (B), I reverse the role of the buyer and seller above. Namely, let seller k be the worst-ranked seller for whom all supplied units have a cost lower than or equal to the values of correspondingly ranked demand units, let buyer $j(k)$ be the counterpart buyer for k 's last unit, and implement rules analogous to those of auction version (A). For each calendar week, I then adopt the version that yields higher trade volume.

A.4 Supplementary descriptions, figures, and tables

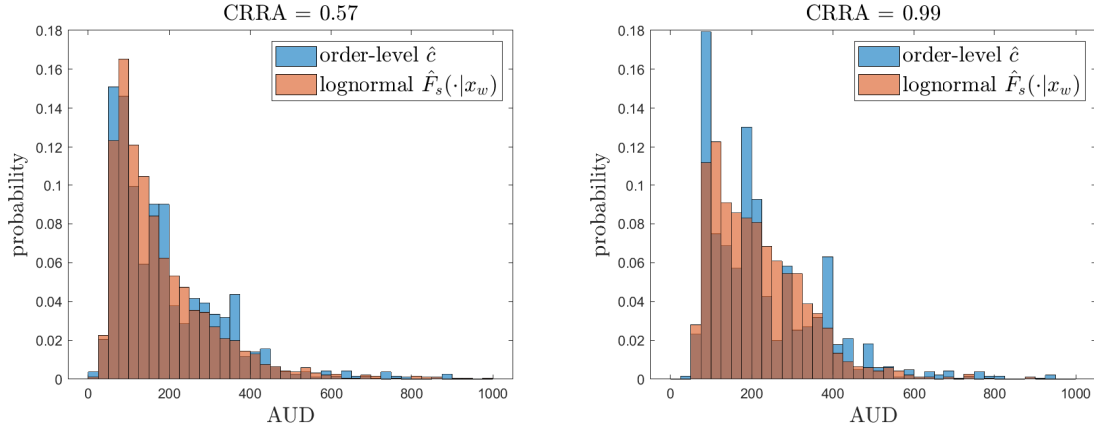
Supplementary figures and tables

Table A1: Estimated parameters of lognormal $\hat{F}_s(c|q, x_w)$ and $\hat{F}_b(v|q, x_w)$, Zone 7

	$\eta = 0.57$				$\eta = 0.99$			
	seller		buyer		seller		buyer	
	$\hat{\theta}_\mu$	$\hat{\theta}_\sigma$	$\hat{\theta}_\mu$	$\hat{\theta}_\sigma$	$\hat{\theta}_\mu$	$\hat{\theta}_\sigma$	$\hat{\theta}_\mu$	$\hat{\theta}_\sigma$
daily storage (log), seasonally adjusted	-2.35	-1.43	-2.40	-0.01	-2.31	-0.25	-2.36	0.95
daily rainfall (mm)	0.02	0.02	-0.01	-0.11	0	-0.01	-0.01	-0.06
log quantity (ML)	0.07	-0.16	-0.05	0.01	0.02	-0.10	-0.06	0
waterday/366								
Bernstein polynomial term 1	0.60	1.01	1.17	0.04	0.44	-1.55	0.77	0.01
Bernstein polynomial term 2	-2.16	0.06	-1.77	0.16	-1.49	1.64	-1.27	1.80
Bernstein polynomial term 3	-0.70	-0.10	-0.40	0.04	-0.72	-0.84	-0.49	-0.88
constant	4.88	-0.96	5.44	-1.68	5.17	-0.97	5.44	-1.43

Notes: θ_μ and θ_σ are parameters of the distribution $\text{Lognormal}(x_w\theta_\mu, x_w\theta_\sigma)$, where $x_w\theta_\mu$ and $x_w\theta_\sigma$ are the mean and standard deviation of $\ln(v)$ for buyers or $\ln(c)$ for sellers, and x_w is a vector of the covariates listed in the left column.

Figure A1: Histogram of order-level $\hat{c}$ versus lognormal $\hat{F}_s(\cdot|q, x_w)$, Zone 7



Notes: These figures compare the histogram of sellers' estimated order-level costs for limit orders, $\hat{c}$, to a histogram of random draws from the estimated lognormal distribution $\hat{F}_s(\cdot|q, x_w)$. The purpose is to check whether the lognormal specification is a good approximation. Specifically, for each $\hat{c}_i$, one draw is randomly drawn from the lognormal distribution conditional on the covariates $x_{w,i}$ corresponding to that order. The left figure is conditional on CRRA parameter 0.57, while the right figure is conditional on CRRA parameter 0.99. The figures are not expected to be identical because $\hat{F}_s(\cdot|q, x_w)$ also includes costs of sellers that placed market orders.

Matching orders and trades

As explained in Section 2.3, the data does not provide links between orders and trades; I infer the linkages by matching the price, volume, and dates of the orders and trades. The set of trades reported by the exchange include, without distinction, both trades from the exchange and external trades that only use its form lodgment services. So only a subset of reported trades will be matched to exchange orders. Here I describe the technical details of the matching process.

I define a trade as a *potential* match for an order if it has the same price and volume in megaliters as the order and has a trade approval date within one week of the last-seen time of the order. The one week window accounts for trades not always being approved instantaneously. I also account for buy limit order prices in this data reporting flat-fee-inclusive prices (reflecting the buyer’s perspective of what they must pay) while sell limit order prices and trade prices reflect pure water prices. Regarding zones, the exchange trade data report the source zone but not the destination zone of trades; nonetheless, the zone composition of a potential match must obey trade restrictions and be corroborated by the Victorian Water Register, which reports the zone composition of trades, though not the platform where the trade was arranged.

To have a systematic method of matching trades to orders given the possibility of multiple potential matches, I apply the following matching algorithm. For any order-trade pair, I define $d = (\text{approval date} - \text{trade date})$ if they are a potential match as defined above and $d = \infty$ otherwise. Each order then has a rank order list of all trades, lowest d first, and the same goes for each trade. I then run the Gale-Shapley Algorithm using these rank order lists, with both trades and orders sorted in descending time order so those with later times “move” first. In the special case of all else being equal, sell limit order matches and same zone matches are prioritized because they reflect the most common trade scenario. After running the algorithm, I also search for cases in which a trade may have been formed not by a market order fulfilling a limit order but by the automatic matching of a buy limit order to a sell limit order with a weakly lower price. I detect less than 10 such cases in the entire data sample. I count all of the limit orders involved in these cases as executed.

Section 6.1 benchmark: additional implementation details

If the supply and demand curves cross at a single point, the market-clearing price and quantity are immediately determined by that point. It is also possible that the curves cross along a vertical or horizontal line due to their steps. In this case, the market-clearing quantity (or total trade volume) is defined such that every unit for which the market-facing value of the buyer is greater than or equal to the market-facing cost of the seller is traded. If the supply and demand curves meet along a vertical line or do not cross at all, then the market-clearing price is defined to be the midpoint of the buyer's and seller's market-facing valuations for the marginal traded unit.

Overall, I am careful to avoid inadvertently advantaging the counterfactual simulations over the observed exchange. In particular, I take steps to address the possibility that some observed limit orders could be revisions, where a trader replaces an unexecuted limit order with a new, revised limit order for the same parcel. In this scenario, if the initial order executes in the counterfactual world (unlike in the data), then the revision should disappear since the parcel has already sold. Likewise, if an initial listing and its revision happen in the same week in the data, they should not both count in the same batch for market clearing in the benchmark. Empirically, I flag potential revisions as instances in which a limit order ends unexecuted and subsequently a new limit order appears on the same day with the same quantity and a different price. In the counterfactuals, I remove all orders flagged as potential revisions in the scenarios above, though not all will be true revisions. By reducing supply and demand in the counterfactuals, these precautionary steps work against the counterfactual exchanges in a comparison versus the observed exchange.

A.5 Counterfactual analysis: Zone 1A results

Table A2: Percent change of counterfactual outcomes from observed exchange, Zone 1A

	CRRA η	counterfactual	total trade surplus	surplus per unit traded	total trade volume
<i>Baseline model</i>					
(a)	0.99	benchmark	+11%	+17%	-5%
(b)	0.57	benchmark	+29%	-18%	+57%
(c)	0.57	dominant strategy double auction	+21%	-12%	+38%
<i>Endogenous quantity model</i>					
(d)	0.99	benchmark	+30%	+44%	-10%
(e)	0.57	benchmark	+45%	-6%	+53%

Notes: Table reports the percent change in simulated counterfactual outcomes from corresponding outcomes of the observed exchange. The product is water sourced from Zone 1A. The time period is the full sample period excluding times when the Goulburn to Murray trade limit was open. Section 6.5 analyzes this trade limit and interzone trade.

Table A3: Percent change of benchmark outcomes from observed exchange, Zone 1A

time period	(a) total trade surplus	(b) surplus per unit traded	(c) total trade volume
<i>Baseline model, CRRA $\eta = 0.99$</i>			
1. Apr 1, 2020–May 30, 2020	+45%	+29%	+12%
2. Jul 1, 2020–Oct 3, 2020	+16%	+14%	+1%
3. Oct 4, 2020–Jan 30, 2021	+7%	+26%	-15%
5. Jul 1, 2021–Oct 2, 2021	+5%	-9%	+15%
<i>Baseline model, CRRA $\eta = 0.57$</i>			
1. Apr 1, 2020–May 30, 2020	+87%	-21%	+136%
2. Jul 1, 2020–Oct 3, 2020	+36%	-31%	+96%
3. Oct 4, 2020–Jan 30, 2021	+16%	-13%	+32%
5. Jul 1, 2021–Oct 2, 2021	+35%	-22%	+74%

Notes: Table reports the percent change in simulated counterfactual outcomes from corresponding outcomes of the observed exchange. The product is water sourced from Zone 1A.

Table A4: Percent change from observed exchange, total trade surplus, Zone 1A

	$\eta = 0.99$	$\eta = 0.57$
(a) observed exchange	0%	0%
<i>Counterfactual simulations, baseline model:</i>		
(b) + parcel splitting (price as observed)	+5%	+11%
(c) + parcel splitting + no price shading	–	+15%
(d) + parcel splitting + no price shading + batching = benchmark	+11%	+29%

Notes: Table reports the percent change in total trade surplus of each simulated counterfactual from that of the observed exchange. The product is water sourced from Zone 1A. The time period is the full sample period excluding times when the Goulburn-to-Murray trade limit was open.